

Solved selected problems of Introductory functional analysis with applications - Erwin Kreyszig

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Chapter 3 - Inner Product Spaces. Hilbert Spaces

3.1 - Inner Product Space. Hilbert Space

Proof. 1 Let us prove (4) as follows

$$\begin{aligned}\|x + y\|^2 + \|x - y\|^2 &= \langle x + y, x + y \rangle + \langle x - y, x - y \rangle \\ &= \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle + \langle x, x \rangle - \langle x, y \rangle - \langle y, x \rangle + \langle y, y \rangle \\ &= 2 \langle x, x \rangle + 2 \langle y, y \rangle \\ &= 2(\|x\|^2 + \|y\|^2)\end{aligned}$$

Where we used the definition of the norm $\|x\| = \sqrt{\langle x, x \rangle}$ and the distributive property of the inner products. $\square$

Proof. 2 Let $x \perp y$ in an inner product space X then

$$\begin{aligned}\|x + y\|^2 &= \langle x + y, x + y \rangle \\ &= \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle \\ &= \langle x, x \rangle + \langle y, y \rangle \\ &= \|x\|^2 + \|y\|^2\end{aligned}$$

Where we used that $\langle x, y \rangle = \langle y, x \rangle = 0$ since $x \perp y$.

Suppose now we have m mutually orthogonal vectors, i.e. $x_1 \perp x_2 \perp \dots \perp x_m$ then

$$\begin{aligned}\|x_1 + x_2 + \dots + x_m\|^2 &= \langle x_1 + x_2 + \dots + x_m, x_1 + x_2 + \dots + x_m \rangle \\ &= \langle x_1, x_1 \rangle + \langle x_1, x_2 \rangle + \dots + \langle x_m, x_{m-1} \rangle + \langle x_m, x_m \rangle \\ &= \langle x_1, x_1 \rangle + \langle x_2, x_2 \rangle + \dots + \langle x_m, x_m \rangle \\ &= \|x_1\|^2 + \|x_2\|^2 + \dots + \|x_m\|^2\end{aligned}$$

Where we used that $\langle x_i, x_j \rangle = 0$ for $i \neq j$ since they are orthogonal. $\square$

Proof. **3** Let $x, y \in \mathbb{R}$ and suppose we have the following property

$$\|x + y\|^2 = \|x\|^2 + \|y\|^2$$

then since $\|x\|^2 = \langle x, x \rangle$ above equation implies that

$$\begin{aligned}\langle x + y, x + y \rangle &= \langle x, x \rangle + \langle y, y \rangle \\ \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle &= \langle x, x \rangle + \langle y, y \rangle \\ \langle x, y \rangle + \langle y, x \rangle &= 0\end{aligned}$$

But since $x, y \in \mathbb{R}$ we have that $\langle x, y \rangle = \langle y, x \rangle$, hence

$$\langle x, y \rangle = \langle y, x \rangle = 0$$

Therefore $x \perp y$.

Let now $x, y \in \mathbb{C}$ and suppose that the property holds in this case too. Then as before must be that

$$\langle x, y \rangle + \langle y, x \rangle = 0$$

Also, we know that $\langle y, x \rangle = \overline{\langle x, y \rangle}$ hence we need that

$$\langle x, y \rangle = -\overline{\langle x, y \rangle}$$

But since we are assuming a complex inner product we have that $\langle x, y \rangle$ is of the form $a + bi$ hence

$$a + bi = -(a - bi) = -a + bi$$

Equating the real and imaginary parts we see that $a = 0$ and $b = b$. So for example if $x = 1 + i$ and $y = 1 - i$ we see that the Pythagorean property holds, as we show below

$$\|1 + i + 1 - i\|^2 = \|2\|^2 = 4 = 2 + 2 = \|1 + i\|^2 + \|1 - i\|^2$$

But inner product gives us

$$\langle x, y \rangle = (1 + i)\overline{(1 - i)} = (1 + i)(1 + i) = 1 + i + i - 1 = 2i$$

Therefore x and y are not orthogonal. □

Proof. **6** Let $x \neq 0$ and $y \neq 0$

- (a) Let $x \perp y$ we want to prove that $\{x, y\}$ is a linearly independent set i.e. we want to prove

$$\alpha_1 x + \alpha_2 y = 0$$

Only for $\alpha_1 = \alpha_2 = 0$. Suppose the opposite we want to arrive at a contradiction, let $\alpha_1 \neq 0$ and $\alpha_2 = 0$ then $\alpha_1 x + 0y = 0$ i.e. $x = 0$ but we said that $x \neq 0$ so it must be that $\alpha_1 \neq 0$ and $\alpha_2 \neq 0$. Then we can write that

$$x = -\frac{\alpha_2}{\alpha_1}y$$

Also, we know that $\langle x, y \rangle = 0$ since $x \perp y$ then

$$\langle x, y \rangle = \left\langle -\frac{\alpha_2}{\alpha_1}y, y \right\rangle = -\frac{\alpha_2}{\alpha_1} \langle y, y \rangle = 0$$

But this implies that $y = 0$ and we said that $y \neq 0$, a contradiction.

Therefore must be that $\{x, y\}$ is a linearly independent set.

- (b) Now, let $\{x_1, \dots, x_m\}$ be a set of mutually orthogonal non-zero vectors, we want to show they are a linearly independent set.

As before, suppose the opposite, we want to arrive at a contradiction.

If $\alpha_i \neq 0$ for only the i th vector then $x_i = 0$ and that cannot happen.

So let us assume that $\alpha_i \neq 0$ for $1 \leq i \leq m$, if not all α_i s were different from zero the following still applies, then

$$\begin{aligned} \alpha_1 x_1 + \dots + \alpha_n x_n &= 0 \\ x_1 &= -\frac{\alpha_2}{\alpha_1}x_2 - \dots - \frac{\alpha_n}{\alpha_1}x_n \end{aligned}$$

Also we know that $\langle x_1, x_2 \rangle = 0$ then

$$\begin{aligned} \langle x_1, x_2 \rangle &= \left\langle -\frac{\alpha_2}{\alpha_1}x_2 - \dots - \frac{\alpha_n}{\alpha_1}x_n, x_2 \right\rangle \\ &= -\frac{\alpha_2}{\alpha_1} \langle x_2, x_2 \rangle - \dots - \frac{\alpha_n}{\alpha_1} \langle x_n, x_2 \rangle \\ &= -\frac{\alpha_2}{\alpha_1} \langle x_2, x_2 \rangle \\ &= 0 \end{aligned}$$

Where we used that $\langle x_i, x_2 \rangle = 0$ for $1 \leq i \leq m$ and $i \neq 2$. But the last equality implies that $x_2 = 0$ which is a contradiction.

Therefore must be that the orthogonal nonzero vectors $\{x_1, \dots, x_m\}$ are a linearly independent set.

□

Proof. 7 Let $\langle x, u \rangle = \langle x, v \rangle$ for all x , then must also be true for $x = u$ i.e. $\langle u, u \rangle = \langle u, v \rangle$ and for $x = v$ i.e. $\langle v, u \rangle = \langle v, v \rangle$.
Now, let us compute $\langle u - v, u - v \rangle$ as follows

$$\begin{aligned}\langle u - v, u - v \rangle &= \langle u, u \rangle - \langle u, v \rangle - \langle v, u \rangle + \langle v, v \rangle \\ &= \langle u, v \rangle - \langle u, v \rangle - \langle v, u \rangle + \langle v, u \rangle \\ &= 0\end{aligned}$$

But then this implies that $u - v = 0$. Therefore $u = v$. $\square$

Proof. 8 Let us compute (9) knowing that $\|x\| = \sqrt{\langle x, x \rangle}$ then

$$\begin{aligned}\frac{1}{4}(\|x + y\|^2 - \|x - y\|^2) &= \frac{1}{4}(\langle x + y, x + y \rangle - \langle x - y, x - y \rangle) \\ &= \frac{1}{4}(\langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle - \\ &\quad - (\langle x, x \rangle + \langle x, -y \rangle + \langle -y, x \rangle + \langle -y, -y \rangle)) \\ &= \frac{1}{4}(\langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle - \\ &\quad - \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle - \langle y, y \rangle) \\ &= \frac{1}{4}(2\langle x, y \rangle + 2\langle y, x \rangle) \\ &= \langle x, y \rangle\end{aligned}$$

Where we used that $\langle x, y \rangle = \langle y, x \rangle$ since we are in a real inner product space. $\square$

Proof. 9 Let us compute (10) knowing that we are in a complex inner product space, then

$$\begin{aligned}
\frac{1}{4}(\|x+y\|^2 - \|x-y\|^2) &= \frac{1}{4}(\langle x+y, x+y \rangle - \langle x-y, x-y \rangle) \\
&= \frac{1}{4}(2\langle x, y \rangle + 2\langle y, x \rangle) \\
&= \frac{1}{4}(2\langle x, y \rangle + 2\overline{\langle x, y \rangle}) \\
&= \frac{1}{2}(\operatorname{Re}\langle x, y \rangle + i\operatorname{Im}\langle x, y \rangle + \operatorname{Re}\langle x, y \rangle - i\operatorname{Im}\langle x, y \rangle) \\
&= \operatorname{Re}\langle x, y \rangle
\end{aligned}$$

Now, we compute the second equation as follows

$$\begin{aligned}
\frac{1}{4}(\|x+iy\|^2 - \|x-iy\|^2) &= \frac{1}{4}(\langle x+iy, x+iy \rangle - \langle x-iy, x-iy \rangle) \\
&= \frac{1}{4}(\langle x, x \rangle + \langle x, iy \rangle + \langle iy, x \rangle + \langle iy, iy \rangle - \\
&\quad - (\langle x, x \rangle + \langle x, -iy \rangle + \langle -iy, x \rangle + \langle -iy, -iy \rangle)) \\
&= \frac{1}{4}(\langle x, x \rangle - i\langle x, y \rangle + i\langle y, x \rangle + \langle y, y \rangle - \\
&\quad - \langle x, x \rangle - i\langle x, y \rangle + i\langle y, x \rangle - \langle y, y \rangle) \\
&= \frac{1}{2}(-i\langle x, y \rangle + i\langle y, x \rangle) \\
&= \frac{1}{2}(-i\langle x, y \rangle + i\overline{\langle x, y \rangle}) \\
&= \frac{1}{2}(-i\operatorname{Re}\langle x, y \rangle + \operatorname{Im}\langle x, y \rangle + i\operatorname{Re}\langle x, y \rangle + \operatorname{Im}\langle x, y \rangle) \\
&= \operatorname{Im}\langle x, y \rangle
\end{aligned}$$

□

Proof. 10 Let z_1 and z_2 be complex numbers and let $\langle z_1, z_2 \rangle = z_1 \overline{z_2}$ we want to prove that in fact this is a valid definition of an inner product. So if this expression defines an inner product, it must have the four properties of inner products. Then for (IP1) let z_3 be also a complex number

$$\langle z_1 + z_3, z_2 \rangle = (z_1 + z_3) \overline{z_2} = z_1 \overline{z_2} + z_3 \overline{z_2} = \langle z_1, z_2 \rangle + \langle z_3, z_2 \rangle$$

For (IP2) we see that

$$\langle \alpha z_1, z_2 \rangle = \alpha z_1 \overline{z_2} = \alpha \langle z_1, z_2 \rangle$$

For (IP3) we see that

$$\langle z_1, z_2 \rangle = z_1 \overline{z_2} = \overline{\overline{z_1} z_2} = \overline{\langle z_2, z_1 \rangle}$$

Finally for (IP4) we see that $\langle z_1, z_1 \rangle = z_1 \overline{z_1} \geq 0$. Also, let $\langle z_1, z_1 \rangle = 0$ then

$$z_1 \overline{z_1} = (x + iy)(x - iy) = x^2 + y^2 = 0$$

Then the only way for this equation to hold is that $x = y = 0$ and hence $z_1 = 0$. Now, if we suppose $z_1 = 0$ then $\langle z_1, z_1 \rangle = z_1 \overline{z_1} = 0$ and hence (IP4) is also satisfied.

Therefore $\langle z_1, z_1 \rangle = z_1 \overline{z_1}$ defines in fact an inner product.

Also, to have orthogonality it must happen that $\langle z_1, z_2 \rangle = z_1 \overline{z_2} = 0$. $\square$

Proof. 11 Let X be the vector space of all ordered pairs of complex numbers. The norm defined as $\|x\| = |\xi_1| + |\xi_2|$ cannot be obtained from an inner product. We prove this by showing that the norm does not satisfy the parallelogram equality. Let us take $x = (-1, 1)$ and $y = (-1, -1)$ then

$$2(\|x\|^2 + \|y\|^2) = 2((|-1| + |1|)^2 + (|-1| + |-1|)^2) = 16$$

But

$$\|x + y\|^2 + \|x - y\|^2 = (|-1 - 1| + |1 - 1|)^2 + (|-1 + 1| + |1 + 1|)^2 = 8$$

Therefore the parallelogram equality is not satisfied and hence the norm defined cannot be obtained from an inner product. $\square$

Proof. **13** The inner product defined in 3.1-5 for continuous functions states that

$$\langle x, y \rangle = \int_a^b x(t) \overline{y(t)} dt$$

Let us check that this definition satisfies (IP1) to (IP4). We assume $x(t)$ and $y(t)$ complex-valued functions.

(IP1):

$$\begin{aligned} \langle x + y, z \rangle &= \int_a^b (x(t) + y(t)) \overline{z(t)} dt \\ &= \int_a^b x(t) \overline{z(t)} dt + \int_a^b y(t) \overline{z(t)} dt \\ &= \langle x, z \rangle + \langle y, z \rangle \end{aligned}$$

(IP2):

$$\langle \alpha x, y \rangle = \int_a^b \alpha x(t) \overline{y(t)} dt = \alpha \int_a^b x(t) \overline{y(t)} dt = \alpha \langle x, y \rangle$$

(IP3):

$$\langle x, y \rangle = \int_a^b x(t) \overline{y(t)} dt = \int_a^b \overline{\overline{x(t)} y(t)} dt = \overline{\langle y, x \rangle}$$

(IP4):

$$\langle x, x \rangle = \int_a^b x(t) \overline{x(t)} dt = \int_a^b |x(t)|^2 dt \geq 0$$

Let now $\langle x, x \rangle = 0$ then

$$\int_a^b |x(t)|^2 dt = 0$$

Then must be that $x(t) = 0$ since $x(t)$ is continuous. Conversely, if we suppose that $x(t) = 0$ we get that

$$\langle x, x \rangle = \int_a^b 0 \cdot 0 dt = 0$$

Therefore the inner product defined in 3.1-5 satisfy the properties (IP1) to (IP4). $\square$

3.2 - Further Properties of Inner Product Spaces

Proof. **2** An example of subspace of l^2 is the set sequences with finitely many entries different from zero followed by zeroes i.e.

$$Y = \{(x_1, x_2, \dots, x_k, 0, 0, 0, \dots) : x_i \in \mathbb{R}, 0 \leq i \leq k\}$$

Suppose $x = (x_1, x_2, \dots, x_k, 0, 0, 0, \dots) \in Y$ then

$$\|x\|_2 = \left(\sum_{i=1}^{\infty} |x_i|^2 \right)^{1/2} = \left(\sum_{i=1}^k |x_i|^2 \right)^{1/2} < \infty$$

So $Y \subset l^2$.

On the other hand, let $x' = (x'_1, x'_2, \dots, x'_k, 0, 0, 0, \dots) \in Y$, and α, β be scalars then

$$\begin{aligned} \alpha x + \beta x' &= \alpha(x_1, x_2, \dots, x_k, 0, 0, 0, \dots) + \beta(x'_1, x'_2, \dots, x'_k, 0, 0, 0, \dots) \\ &= (\alpha x_1, \alpha x_2, \dots, \alpha x_k, 0, 0, 0, \dots) + (\beta x'_1, \beta x'_2, \dots, \beta x'_k, 0, 0, 0, \dots) \\ &= (\alpha x_1 + \beta x'_1, \alpha x_2 + \beta x'_2, \dots, \alpha x_k + \beta x'_k, 0, 0, 0, \dots) \end{aligned}$$

We see that $\alpha x_i + \beta x'_i \in \mathbb{R}$ since $\mathbb{R}$ is a vector space then Y is a subspace of l^2 . $\square$

Proof. 3 Let X be the inner product space consisting of the polynomial $x = 0$ and all real polynomials in t of degree not exceeding 2, considered for real $t \in [a, b]$ with inner product defined as

$$\langle x, y \rangle = \int_a^b x(t)y(t) dt$$

Let $(x_n) = (\alpha_n t^2 + \beta_n t + \gamma_n) \in X$ be a Cauchy sequence on X where $\alpha_n, \beta_n, \gamma_n \in \mathbb{R}$ then since this is a Cauchy sequence, given $\epsilon > 0$ we can find $N \in \mathbb{N}$ such that when $n, m > N$ we have that

$$\|x_n - x_m\| = \left(\int_a^b (x_n(t) - x_m(t))^2 dt \right)^{1/2} < \epsilon$$

Hence

$$\left(\int_a^b ((\alpha_n - \alpha_m)t^2 + (\beta_n - \beta_m)t + (\gamma_n - \gamma_m))^2 dt \right)^{1/2} < \epsilon$$

For this integral to tend to 0 over the fixed interval $[a, b]$ since t does not depend on n must happen that $(\alpha_n - \alpha_m) \rightarrow 0$, $(\beta_n - \beta_m) \rightarrow 0$ and $(\gamma_n - \gamma_m) \rightarrow 0$.

This implies that (α_n) , (β_n) and (γ_n) are Cauchy sequences of real numbers which we know they converge, let us say that they converge to α, β and γ respectively.

Now, let us define $x = \alpha t^2 + \beta t + \gamma$ then

$$\begin{aligned} \|x_n - x\| &= \left(\int_a^b (x_n(t) - x(t))^2 dt \right)^{1/2} \\ &= \left(\int_a^b ((\alpha_n - \alpha)t^2 + (\beta_n - \beta)t + (\gamma_n - \gamma))^2 dt \right)^{1/2} \end{aligned}$$

But we know that $\alpha_n \rightarrow \alpha$, $\beta_n \rightarrow \beta$ and $\gamma_n \rightarrow \gamma$ therefore we have that

$$\left(\int_a^b ((\alpha_n - \alpha)t^2 + (\beta_n - \beta)t + (\gamma_n - \gamma))^2 dt \right)^{1/2} \rightarrow 0$$

Or what is the same $x_n \rightarrow x$ and hence since x_n was a Cauchy sequence then X is complete.

Let now Y be all $x \in X$ such that $x(a) = 0$. Let $y, y' \in Y$ and let α, β be scalars. We know that when $t = a$ then $y(a) = y'(a) = 0$ then for $z(t) = \alpha y(t) + \beta y'(t)$ we have that $z(a) = \alpha y(a) + \beta y'(a) = 0 + 0 = 0$. Therefore Y is a subspace of X .

Finally, let Z be all $x \in X$ of degree 2. Let $z, z' \in Z$ and let λ, ρ be scalars. Then let us compute $\lambda z + \rho z'$ as follows

$$\lambda(\alpha t^2 + \beta t + \gamma) + \rho(\alpha' t^2 + \beta' t + \gamma') = (\lambda\alpha + \rho\alpha')t^2 + (\lambda\beta + \rho\beta')t + (\lambda\gamma + \rho\gamma')$$

Where $\alpha, \alpha', \beta, \beta', \gamma$ and γ' are not zero. Then we see that if $\lambda = \rho = 0$ we get that $\lambda z + \rho z' = 0$ but the polynomial $z = 0$ is not included in Z then Z is not a subspace of X . $\square$

Proof. 4 Let $y \perp x_n$ and $x_n \rightarrow x$ then by the Continuity of the Inner Product we have that

$$\langle x_n, y \rangle \rightarrow \langle x, y \rangle$$

This implies that

$$\| \langle x_n, y \rangle - \langle x, y \rangle \| \rightarrow 0$$

But we know that $\langle x_n, y \rangle = 0$ because $y \perp x_n$ hence

$$\| \langle x, y \rangle \| \rightarrow 0$$

And by the (N2) property for norms, we get that

$$\langle x, y \rangle = 0$$

Therefore $x \perp y$. $\square$

Proof. 5 Let a sequence (x_n) in an inner product space such that $\|x_n\| \rightarrow \|x\|$ and $\langle x_n, x \rangle \rightarrow \langle x, x \rangle$. Given that $\|x_n\| \rightarrow \|x\|$ then $\|x_n\|^2 \rightarrow \|x\|^2$ so given $\epsilon^2/3 > 0$ there is $N \in \mathbb{N}$ such that when $n \geq N$ we see that

$$|\|x_n\|^2 - \|x\|^2| = |\langle x_n, x_n \rangle - \langle x, x \rangle| < \epsilon^2/3$$

On the other hand, since $\langle x_n, x \rangle \rightarrow \langle x, x \rangle$ given $\epsilon^2/3 > 0$ there is $N' \in \mathbb{N}$ such that when $n \geq N'$ we get that

$$|\langle x_n, x \rangle - \langle x, x \rangle| = |\langle x_n - x, x \rangle| < \epsilon^2/3$$

Also, let us note that

$$\begin{aligned} \|x_n - x\|^2 &= |\langle x_n - x, x_n - x \rangle| \\ &= |\langle x_n - x, x_n \rangle - \langle x_n - x, x \rangle| \\ &\leq |\langle x_n - x, x_n \rangle| + |\langle x_n - x, x \rangle| \\ &\leq |\langle x_n, x_n \rangle - \langle x, x_n \rangle| + |\langle x_n - x, x \rangle| \\ &\leq |\langle x_n, x_n \rangle - \langle x, x \rangle + \langle x, x \rangle - \langle x, x_n \rangle| + |\langle x_n - x, x \rangle| \\ &\leq |\langle x_n, x_n \rangle - \langle x, x \rangle| + |\langle x, x - x_n \rangle| + |\langle x_n - x, x \rangle| \end{aligned}$$

Then if we take $M = \max(N, N')$ when $n \geq M$ we have that

$$\begin{aligned} \|x_n - x\|^2 &\leq |\langle x_n, x_n \rangle - \langle x, x \rangle| + |\langle x, x - x_n \rangle| + |\langle x_n - x, x \rangle| \\ &< \epsilon^2/3 + \epsilon^2/3 + \epsilon^2/3 = \epsilon^2 \end{aligned}$$

Therefore $\|x_n - x\| < \epsilon$ and hence $x_n \rightarrow x$. □

Proof. 9 Let V be the vector space of all continuous complex-valued functions on $J = [a, b]$. Let $X_1 = (V, \|\cdot\|_\infty)$ where $\|x\|_\infty = \max_{t \in J} |x(t)|$ and let $X_2 = (V, \|\cdot\|_2)$ where

$$\|x\|_2 = \sqrt{\langle x, x \rangle} \quad \langle x, y \rangle = \int_a^b x(t) \overline{y(t)} dt$$

Also, let us define $i : X_1 \rightarrow X_2$ be the identity mapping $x \rightarrow x$. Suppose we take $x, y \in V$ then we see that

$$\begin{aligned} \|x - y\|_2 &= \sqrt{\langle x - y, x - y \rangle} \\ &= \sqrt{\int_a^b (x(t) - y(t)) \overline{(x(t) - y(t))} dt} \\ &= \sqrt{\int_a^b |x(t) - y(t)|^2 dt} \end{aligned}$$

But also we have that

$$\sqrt{\int_a^b |x(t) - y(t)|^2 dt} \leq \sqrt{\int_a^b \max_{t \in J} |x(t) - y(t)|^2 dt} = \max_{t \in J} |x(t) - y(t)| \sqrt{b - a}$$

So given $\epsilon > 0$ if we take $\delta = \epsilon / \sqrt{b - a}$ such that

$$\|x - y\|_\infty = \max_{t \in J} |x(t) - y(t)| < \delta = \frac{\epsilon}{\sqrt{b - a}}$$

We get that

$$\|x - y\|_2 \leq \max_{t \in J} |x(t) - y(t)| \sqrt{b - a} < \epsilon$$

Therefore the identity mapping i is continuous. □

Proof. 10 Let $T : X \rightarrow X$ be a bounded linear operator on a complex inner product space X and let $\langle Tx, x \rangle = 0$ for all $x \in X$. Let $x = \alpha u + v \in X$ for some $u, v \in X$ and $\alpha \in \mathbb{C}$ then

$$\begin{aligned}\langle Tx, x \rangle &= \langle \alpha Tu + Tv, \alpha u + v \rangle \\ &= \langle \alpha Tu, \alpha u + v \rangle + \langle Tv, \alpha u + v \rangle \\ &= \alpha \bar{\alpha} \langle Tu, u \rangle + \alpha \langle Tu, v \rangle + \bar{\alpha} \langle Tv, u \rangle + \langle Tv, v \rangle \\ &= \alpha \langle Tu, v \rangle + \bar{\alpha} \langle Tv, u \rangle\end{aligned}$$

Then taking $\alpha = 1$ we get that

$$\begin{aligned}\langle Tu, v \rangle + \langle Tv, u \rangle &= 0 \\ \langle Tu, v \rangle &= -\langle Tv, u \rangle\end{aligned}$$

But if we take $\alpha = i$ we have that $i \langle Tu, v \rangle - i \langle Tv, u \rangle = 0$ or

$$\begin{aligned}\langle Tu, v \rangle - \langle Tv, u \rangle &= 0 \\ \langle Tu, v \rangle &= \langle Tv, u \rangle\end{aligned}$$

This implies that $\langle Tu, v \rangle = 0$ for all $u, v \in X$ so in particular if we select $v = Tu$ then $\langle Tu, Tu \rangle = 0$ and hence by (IP4) we have that $Tu = 0$. Therefore $T = 0$.

Let us consider now the Euclidean plane, a real inner product space, and let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by

$$T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \cos \pi/2 & -\sin \pi/2 \\ \sin \pi/2 & \cos \pi/2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -y \\ x \end{bmatrix}$$

Then let us compute the following inner product

$$\left\langle T \begin{bmatrix} x \\ y \end{bmatrix}, \begin{bmatrix} x \\ y \end{bmatrix} \right\rangle = \left\langle \begin{bmatrix} -y \\ x \end{bmatrix}, \begin{bmatrix} x \\ y \end{bmatrix} \right\rangle = -yx + xy = 0$$

Therefore, we see that the inner product is 0 but $T \neq 0$ so above result does not hold for real inner product spaces. $\square$

3.3 - Orthogonal Complements and Direct Sums

Proof. 1 Let H be a Hilbert space, $M \subset H$ a convex subset, and (x_n) a sequence in M such that $\|x_n\| \rightarrow d$ where $d = \inf_{x \in M} \|x\|$.

Let us consider $\overline{M}$ the closure of M then $\overline{M}$ is complete because $\overline{M} \subset H$, also, $(x_n) \in \overline{M}$.

Let $n, m \in \mathbb{N}$ where $n \geq m$. We see that $\frac{1}{2}x_n + \frac{1}{2}x_m \in M$ because M is convex but also we have that

$$\left\| \frac{x_n + x_m}{2} \right\| \geq d = \inf_{x \in M} \|x\|$$

Hence

$$\|x_n + x_m\| \geq 2d$$

On the other hand, by the parallelogram equality we get that

$$\begin{aligned} \|x_n - x_m\|^2 &= -\|x_n + x_m\|^2 + 2(\|x_n\|^2 + \|x_m\|^2) \\ &\leq -(2d)^2 + 2(\|x_n\|^2 + \|x_m\|^2) \end{aligned}$$

But since $\|x_n\| \rightarrow d$ as $n, m \rightarrow \infty$ then $\|x_n - x_m\| \rightarrow 0$. Therefore (x_n) is Cauchy, but this also implies that (x_n) converges because $\overline{M}$ is a subset of a Hilbert space. $\square$

Proof. **5** Let $X = \mathbb{R}^2$ then

- (a) Let $M = \{x\}$ where $x = (\xi_1, \xi_2) \neq 0$ then $M^\perp$ by definition is

$$M^\perp = \{x' \in X \mid x' \perp M\} = \{x' \in X \mid \langle x', x \rangle = 0\}$$

Let $x' = (-\xi_2, \xi_1) \in X$ then we see that

$$\langle x', x \rangle = -\xi_1\xi_2 + \xi_2\xi_1 = 0$$

So $(-\xi_2, \xi_1) \in M^\perp$ but $(-\alpha\xi_2, \alpha\xi_1)$ for $\alpha \in \mathbb{R}$ is also in $M^\perp$.

Therefore $M^\perp = \{(-\alpha\xi_2, \alpha\xi_1) \in X \mid \alpha \in \mathbb{R}\}$.

- (b) Let $M = \{x_1, x_2\}$ where x_1 and x_2 are linearly independent then

$$\begin{aligned} M^\perp &= \{x' \in X \mid x' \perp M\} \\ &= \{x' \in X \mid \langle x', x_1 \rangle = 0 \text{ and } \langle x', x_2 \rangle = 0\} \end{aligned}$$

Suppose $x' \in X$ satisfy $\langle x', x_1 \rangle = 0$ and $\langle x', x_2 \rangle = 0$, then we have that

$$\langle x', x_1 \rangle = \eta_1\xi_{11} + \eta_2\xi_{12} = 0$$

and

$$\langle x', x_2 \rangle = \eta_1\xi_{21} + \eta_2\xi_{22} = 0$$

Where we assumed that $x_1 = (\xi_{11}, \xi_{12})$, $x_2 = (\xi_{21}, \xi_{22})$ and $x' = (\eta_1, \eta_2)$. But since x_1 and x_2 are linearly independent then the equations

$$c_1\xi_{11} + c_2\xi_{21} = 0$$

$$c_1\xi_{12} + c_2\xi_{22} = 0$$

have only one solution, $c_1 = c_2 = 0$.

Therefore must be that $x' = 0$ and hence

$$M^\perp = \{0\}$$

□

Proof. **6**

Let $Y = \{x \mid x = (\xi_j) \in l^2, \xi_{2n} = 0, n \in \mathbb{N}\}$ we want to show that Y is a closed subspace of l^2 .

Let $(x_j) \subset Y$ and suppose it converges to some $x = (\eta_j) \notin Y$ we want to arrive at a contradiction, then given $\epsilon > 0$ there is some $N \in \mathbb{N}$ such that when $n \geq N$ we have that $\|x_n - x\| < \epsilon$, also, we see that

$$\|x_n - x\| = \sqrt{\sum_{j=1}^{\infty} |\xi_j - \eta_j|^2} = \sqrt{\sum_{j=1}^{\infty} |\xi_{2j+1} - \eta_{2j+1}|^2 + \sum_{j=1}^{\infty} |\eta_{2j}|^2}$$

Since the first term on the RHS is non-negative then

$$\|x_n - x\| \geq \sqrt{\sum_{j=1}^{\infty} |\eta_{2j}|^2}$$

So we have a lower bound on $\|x_n - x\|$ but $\|x_n - x\| \rightarrow 0$, a contradiction. Therefore, must be that $\eta_{2j} = 0$ so $x \in Y$ and hence Y is closed.

Given $x \in Y$, to find $Y^\perp$ we need to find all elements $z \in l^2$ such that $\langle x, z \rangle = 0$. We see that

$$\langle x, z \rangle = \sum_{j=1}^{\infty} \xi_j \bar{\eta}_j = \sum_{j=1}^{\infty} \xi_{2j+1} \bar{\eta}_{2j+1} + \sum_{j=1}^{\infty} \xi_{2j} \bar{\eta}_{2j} = \sum_{j=1}^{\infty} \xi_{2j+1} \bar{\eta}_{2j+1}$$

So for $\langle x, z \rangle = 0$ then must be that $\bar{\eta}_{2j+1} = 0$ because it must hold for every $x \in Y$. Therefore

$$Y^\perp = \{x \mid x = (\eta_j) \in l^2, \eta_{2n+1} = 0, n \in \mathbb{N}\}$$

Finally, let $Y = \text{span}\{e_1, \dots, e_n\} \subset l^2$ where $e_j = (\delta_{jk})$.

We see that an element $x \in Y$ can be written as

$$x = \alpha_1 e_1 + \dots + \alpha_n e_n = \left(\alpha_1, \alpha_2, \dots, \alpha_n, 0, 0, \dots \right)$$

Where $\alpha_1, \dots, \alpha_n \in \mathbb{C}$.

Let then $z \in l^2$, we see that

$$\langle x, z \rangle = \alpha_1 \bar{\eta}_1 + \alpha_2 \bar{\eta}_2 + \dots + \alpha_n \bar{\eta}_n + 0 \cdot \bar{\eta}_{n+1} + 0 \cdot \bar{\eta}_{n+2} + \dots$$

So for $\langle x, z \rangle = 0$ must be that $\eta_j = 0$ for $1 \leq j \leq n$ and hence

$$Y^\perp = \{x \mid x = (\xi_j) \in l^2 \text{ such that } \xi_j = 0 \text{ for } 1 \leq j \leq n\}$$

□

Proof. **7**

Let A and $A \subset B$ be nonempty subsets of an inner product space X

- (a) Let $a \in A$ and $a' \in A^\perp$ we see that $\langle a, a' \rangle = 0$ hence $a \perp A^\perp$, also by definition we know that

$$A^{\perp\perp} = \{x \in X \mid x \perp A^\perp\}$$

Then $a \in A^{\perp\perp}$ and since a was arbitrary we have that

$$A \subset A^{\perp\perp}$$

- (b) Let $b \in B$, $a \in A$ and $b' \in B^\perp$, by definition $b' \perp B$ hence $\langle b, b' \rangle = 0$ but since a is also in B then must be that $\langle a, b' \rangle = 0$ and since a and b were arbitrary then this implies that $b' \perp A$ and therefore

$$B^\perp \subset A^\perp$$

- (c) From part (a) we have that $A \subset A^{\perp\perp}$ and from part (b) taking $B = A^{\perp\perp}$ we have that

$$B^\perp = A^{\perp\perp\perp} \subset A^\perp$$

Also, from part (a) if we take A as $A^\perp$ following the same steps we get that

$$A^\perp \subset A^{\perp\perp\perp}$$

Therefore joining both results we get that $A^{\perp\perp\perp} = A^\perp$.

□

Proof. **9**

Let Y be a subspace of a Hilbert space H

($\Rightarrow$) Suppose Y is closed in H then by Lemma 3.3-6 we get that $Y = Y^{\perp\perp}$.

($\Leftarrow$) Suppose $Y = Y^{\perp\perp}$. We prove first that $Y^\perp$ is closed in H .

Let $(x_n) \in Y^\perp$ be a convergent sequence, that converges to $x \in H$.

Let also, $y \in Y$ then we know that $\langle x_n, y \rangle = 0$, then by continuity of the inner product we have that

$$\langle x_n, y \rangle \rightarrow \langle x, y \rangle$$

but $\langle x_n, y \rangle = 0$ then $\langle x, y \rangle = 0$ and hence $x \in Y^\perp$. So $Y^\perp$ is closed.

We see that $Y^\perp$ is a subspace of H since if $y'_1, y'_2 \in Y^\perp$ and $x \in Y$ then

$$\langle \alpha y'_1 + \beta y'_2, x \rangle = \alpha \langle y'_1, x \rangle + \beta \langle y'_2, x \rangle = 0$$

So following the same logic we used to prove $Y^\perp$ is closed assuming Y is a subspace of H we get that $Y^{\perp\perp}$ is closed but $Y = Y^{\perp\perp}$.

Therefore Y is closed.

□

Proof. **10**

Let $M \neq \emptyset$ be any subset of a Hilbert space H .

Let us notice that $M^{\perp\perp}$ is a closed subspace of H so $M^{\perp\perp}$ is a Hilbert space and hence without loss of generality we may assume that M is a subset of $H = M^{\perp\perp}$.

Let us take $x, y \in H = M^{\perp\perp}$ then $\langle x, y \rangle \neq 0$ because otherwise either x or y are not in $M^{\perp\perp}$ so the only way where $\langle x, y \rangle = 0$ is that $x = y = 0$ and hence $M^\perp = \{0\}$.

Then by theorem 3.3-7 we have that the span of M is dense in $M^{\perp\perp}$ but then

$$M \subset \overline{\text{span}(M)} = M^{\perp\perp}$$

Therefore $M^{\perp\perp}$ is the smallest closed subspace containing M . □

3.4 - Orthonormal Sets and Sequences

Proof. 1. Let X be an inner product space of finite dimension n , then, X contains a linearly independent set of n vectors (a basis), let us call them $B' = \{b'_1, \dots, b'_n\}$.

We want to define $B = \{b_1, \dots, b_n\}$ such that the elements are orthonormal. Let us define $b_1 \in X$ such that

$$b_1 = \frac{1}{\|b'_1\|} \cdot b'_1$$

We see that we can write b'_2 as

$$b'_2 = \langle b'_2, b_1 \rangle b_1 + v_2$$

where v_2 is non-zero since the elements of B' are linearly independent, then

$$v_2 = b'_2 - \langle b'_2, b_1 \rangle b_1$$

We see that

$$\begin{aligned} \langle b_1, v_2 \rangle &= \langle b_1, b'_2 - \langle b'_2, b_1 \rangle b_1 \rangle \\ &= \langle b_1, b'_2 \rangle - \langle b_1, \langle b'_2, b_1 \rangle b_1 \rangle \\ &= \langle b_1, b'_2 \rangle - \overline{\langle b'_2, b_1 \rangle} \langle b_1, b_1 \rangle \\ &= \langle b_1, b'_2 \rangle - \langle b_1, b'_2 \rangle \frac{1}{\|b'_1\|^2} \langle b'_1, b'_1 \rangle \\ &= \langle b_1, b'_2 \rangle - \langle b_1, b'_2 \rangle \frac{\|b'_1\|^2}{\|b'_1\|^2} \\ &= \langle b_1, b'_2 \rangle - \langle b_1, b'_2 \rangle \\ &= 0 \end{aligned}$$

Hence, we see that they are orthogonal. Then we define b_2 as

$$b_2 = \frac{1}{\|v_2\|} \cdot v_2$$

Now, to define b_3 we note that there is v_3 such that

$$v_3 = b'_3 - \langle b'_3, b_1 \rangle b_1 - \langle b'_3, b_2 \rangle b_2$$

We see that v_3 is non-zero since the elements of B' are linearly independent. Also, $v_3 \perp b_1$ and $v_3 \perp b_2$, then we define b_3 as

$$b_3 = \frac{1}{\|v_3\|} \cdot v_3$$

We can carry on this process n times until we find

$$v_n = b'_n - \sum_{i=1}^{n-1} \langle b'_n, b_i \rangle b_i$$

From which we define b_n as

$$b_n = \frac{1}{\|v_n\|} \cdot v_n$$

Therefore we have found a set B that is orthogonal pairwise, where each element has norm equal to 1, so B is an orthonormal set.

Finally, because of Lemma 3.4-2 we know that B is linearly independent so B is a basis for X as well. $\square$

Proof. **2.** Equation (12*) states that

$$\sum_{k=1}^n |\langle x, e_k \rangle|^2 \leq \|x\|^2$$

Where $x \in \mathbb{R}^r$ and $r \geq n$. We know from equation (9) that

$$\|y\|^2 = \sum_{k=1}^n |\langle x, e_k \rangle|^2$$

So y is the projection of x onto a set of orthonormal vectors in $\mathbb{R}^n$ hence (12*) is telling us geometrically that the length of a projection vector y is less than or equal to the length of x . $\square$

Proof. 3. Let $x, y \in X$ where X is an inner product space. Let us take an orthonormal set $\{e_1, \dots, e_n\}$ such that

$$e_1 = \frac{y}{\|y\|}$$

Then we see that

$$\begin{aligned} \sum_{k=1}^n |\langle x, e_k \rangle|^2 &= |\langle x, e_1 \rangle|^2 + \sum_{k=2}^n |\langle x, e_k \rangle|^2 \\ &= \left| \left\langle x, \frac{y}{\|y\|} \right\rangle \right|^2 + \sum_{k=2}^n |\langle x, e_k \rangle|^2 \\ &= \frac{|\langle x, y \rangle|^2}{\|y\|^2} + \sum_{k=2}^n |\langle x, e_k \rangle|^2 \end{aligned}$$

We note that

$$\frac{|\langle x, y \rangle|^2}{\|y\|^2} \leq \frac{|\langle x, y \rangle|^2}{\|y\|^2} + \sum_{k=2}^n |\langle x, e_k \rangle|^2$$

Now, applying (12*) to x we get that

$$\frac{|\langle x, y \rangle|^2}{\|y\|^2} \leq \frac{|\langle x, y \rangle|^2}{\|y\|^2} + \sum_{k=2}^n |\langle x, e_k \rangle|^2 = \sum_{k=1}^n |\langle x, e_k \rangle|^2 \leq \|x\|^2$$

Hence

$$|\langle x, y \rangle|^2 \leq \|x\|^2 \|y\|^2$$

And taking the square root we get Schwarz inequality

$$|\langle x, y \rangle| \leq \|x\| \|y\|$$

□

Proof. 4. Let us consider the orthonormal sequence

$$\begin{aligned}e_1 &= (0, 1, 0, 0, \dots) \\e_2 &= (0, 0, 1, 0, \dots) \\e_3 &= (0, 0, 0, 1, \dots) \\&\vdots\end{aligned}$$

And let us consider the sequence

$$x = (1, 0, 0, 0, \dots) \in l^2$$

Then we see that

$$\sum_{k=1}^{\infty} |\langle x, e_k \rangle|^2 = 0$$

But

$$\|x\|^2 = 1^2 + 0 + 0 + \dots = 1$$

Therefore the strict inequality holds in this case

$$\sum_{k=1}^{\infty} |\langle x, e_k \rangle|^2 = 0 < \|x\|^2 = 1$$

□

Proof. 5. Let (e_k) be an orthonormal sequence in an inner product space X , let also $x \in X$ and

$$y = \sum_{k=1}^n \alpha_k e_k \quad \alpha_k = \langle x, e_k \rangle$$

We want to show that $x - y$ is orthogonal to the subspace $Y_n = \text{span}\{e_1, \dots, e_n\}$. Let us consider some $y' \in Y_n$ defined as

$$y' = \sum_{k=1}^n \langle y', e_k \rangle e_k$$

Then

$$\begin{aligned} \langle x - y, y' \rangle &= \langle x, y' \rangle - \langle y, y' \rangle \\ &= \left\langle x, \sum_{k=1}^n \langle y', e_k \rangle e_k \right\rangle - \left\langle \sum_{k=1}^n \langle x, e_k \rangle e_k, \sum_{j=1}^n \langle y', e_j \rangle e_j \right\rangle \\ &= \sum_{k=1}^n \overline{\langle y', e_k \rangle} \langle x, e_k \rangle - \sum_{k=1}^n \langle x, e_k \rangle \left\langle e_k, \sum_{j=1}^n \langle y', e_j \rangle e_j \right\rangle \\ &= \sum_{k=1}^n \overline{\langle y', e_k \rangle} \langle x, e_k \rangle - \sum_{k=1}^n \langle x, e_k \rangle \sum_{j=1}^n \overline{\langle y', e_j \rangle} \langle e_k, e_j \rangle \end{aligned}$$

Since $\langle e_k, e_j \rangle = 1$ only if $k = j$ otherwise it's 0 then we can write that

$$\langle x - y, y' \rangle = \sum_{k=1}^n \overline{\langle y', e_k \rangle} \langle x, e_k \rangle - \sum_{k=1}^n \langle x, e_k \rangle \overline{\langle y', e_k \rangle} = 0$$

Therefore since y' was arbitrary we see that $x - y$ is orthogonal to Y_n . $\square$

Proof. 6. Let $\{e_1, \dots, e_n\}$ be an orthonormal set in an inner product space X , where n is fixed. Let $x \in X$ and let $y = \beta_1 e_1 + \dots + \beta_n e_n$. We want to show that $\|x - y\|$ is minimum if and only if $\beta_j = \langle x, e_j \rangle$ where $j = 1, \dots, n$.

($\Rightarrow$) Suppose $\|x - y\|$ is minimum then we note that

$$x - y = x - \sum_{j=1}^n \beta_j e_j = \left(x - \sum_{i=1}^n \langle x, e_i \rangle e_i \right) + \left(\sum_{k=1}^n \langle x, e_k \rangle e_k - \sum_{j=1}^n \beta_j e_j \right)$$

Then we see that

$$\begin{aligned} & \left\langle x - \sum_{i=1}^n \langle x, e_i \rangle e_i, \sum_{k=1}^n \langle x, e_k \rangle e_k - \sum_{j=1}^n \beta_j e_j \right\rangle \\ &= \left\langle x, \sum_{k=1}^n \langle x, e_k \rangle e_k \right\rangle - \left\langle x, \sum_{j=1}^n \beta_j e_j \right\rangle - \left\langle \sum_{i=1}^n \langle x, e_i \rangle e_i, \sum_{k=1}^n \langle x, e_k \rangle e_k \right\rangle \\ & \quad + \left\langle \sum_{i=1}^n \langle x, e_i \rangle e_i, \sum_{j=1}^n \beta_j e_j \right\rangle \\ &= \sum_{k=1}^n \overline{\langle x, e_k \rangle} \langle x, e_k \rangle - \sum_{j=1}^n \overline{\beta_j} \langle x, e_j \rangle - \sum_{i=1}^n \langle x, e_i \rangle \sum_{k=1}^n \overline{\langle x, e_k \rangle} \langle e_i, e_k \rangle \\ & \quad + \sum_{i=1}^n \langle x, e_i \rangle \sum_{j=1}^n \overline{\beta_j} \langle e_i, e_j \rangle \\ &= \sum_{k=1}^n \overline{\langle x, e_k \rangle} \langle x, e_k \rangle - \sum_{j=1}^n \overline{\beta_j} \langle x, e_j \rangle - \sum_{i=1}^n \langle x, e_i \rangle \overline{\langle x, e_i \rangle} + \sum_{i=1}^n \langle x, e_i \rangle \overline{\beta_i} \\ &= 0 \end{aligned}$$

So the vectors are orthogonal, then by the Pythagorean relation we have that

$$\|x - \sum_{j=1}^n \beta_j e_j\|^2 = \|x - \sum_{i=1}^n \langle x, e_i \rangle e_i\|^2 + \|\sum_{k=1}^n \langle x, e_k \rangle e_k - \sum_{j=1}^n \beta_j e_j\|^2$$

We see the last term is

$$\begin{aligned} \left\| \sum_{k=1}^n (\langle x, e_k \rangle - \beta_k) e_k \right\|^2 &= \left\langle \sum_{k=1}^n (\langle x, e_k \rangle - \beta_k) e_k, \sum_{j=1}^n (\langle x, e_j \rangle - \beta_j) e_j \right\rangle \\ &= \sum_{k=1}^n (\langle x, e_k \rangle - \beta_k) \sum_{j=1}^n (\langle x, e_j \rangle - \beta_j) \langle e_k, e_j \rangle \\ &= \sum_{k=1}^n |\langle x, e_k \rangle - \beta_k|^2 \end{aligned}$$

Therefore since the first term, $\|x - \sum_{i=1}^n \langle x, e_i \rangle e_i\|^2$, is constant, and we said that $\|x - y\|^2$ is minimum then must be that

$$\sum_{k=1}^n |\langle x, e_k \rangle - \beta_k|^2 = 0$$

And that can only happen if $\beta_k = \langle x, e_k \rangle$ for all $k \in \{1, \dots, n\}$.

($\Leftarrow$) Suppose now that $\beta_j = \langle x, e_j \rangle$ for all $j \in \{1, \dots, n\}$ then

$$\|x - \sum_{j=1}^n \beta_j e_j\|^2 = \|x - \sum_{i=1}^n \langle x, e_i \rangle e_i\|^2 + \sum_{j=1}^n |\langle x, e_j \rangle - \beta_j|^2 = \|x - \sum_{i=1}^n \langle x, e_i \rangle e_i\|^2$$

Since $\|x - \sum_{i=1}^n \langle x, e_i \rangle e_i\|^2$ is a constant then $\|x - y\|$ cannot take a smaller value, i.e. it's the minimum when $\beta_j = \langle x, e_j \rangle$.

□

Proof. 7. Let (e_j) be any orthonormal sequence in an inner product space X and let $x, y \in X$.

Let us consider the Cauchy-Schwartz inequality over the sequences $(\langle x, e_k \rangle)$ and $(\langle y, e_k \rangle)$, then

$$\sum_{k=1}^{\infty} |\langle x, e_k \rangle \langle y, e_k \rangle| \leq \sqrt{\sum_{j=1}^{\infty} |\langle x, e_j \rangle|^2} \sqrt{\sum_{m=1}^{\infty} |\langle y, e_m \rangle|^2}$$

We know that $\sum_{j=1}^{\infty} |\langle x, e_j \rangle|^2 \leq \|x\|^2$ because of (12) hence

$$\sum_{k=1}^{\infty} |\langle x, e_k \rangle \langle y, e_k \rangle| \leq \sqrt{\|x\|^2} \sqrt{\|y\|^2} = \|x\| \|y\|$$

□

Proof. 8. Let $x \in X$ where X is an inner product space and let n_m be the number of $\langle x, e_k \rangle$ such that $|\langle x, e_k \rangle| > 1/m$. From equation (12) we know that

$$\sum_{k=1}^{\infty} |\langle x, e_k \rangle|^2 \leq \|x\|^2$$

So we see that

$$\frac{n_m}{m^2} < \sum_{k=1}^{\infty} |\langle x, e_k \rangle|^2 \leq \|x\|^2$$

Therefore, must be that

$$n_m < m^2 \|x\|^2$$

□

Proof. 9. Let the sequence $(x_0, x_1, x_2, \dots)$ where $x_j(t) = t^j$ on the interval $[-1, 1]$ where

$$\langle x, y \rangle = \int_{-1}^1 x(t)y(t) dt$$

To orthonormalize the first three terms of the sequence we follow the Gram-Schmidt process. Let us compute first $\|x_0\|$ as follows

$$\|x_0\| = \sqrt{\langle x_0, x_0 \rangle} = \sqrt{\int_{-1}^1 x_0(t)x_0(t) dt} = \sqrt{\int_{-1}^1 t^0 t^0 dt} = \sqrt{\int_{-1}^1 dt} = \sqrt{2}$$

So the first element of our orthonormalized sequence is

$$e_0 = \frac{1}{\|x_0\|} x_0 = \frac{1}{\sqrt{2}} \cdot t^0 = \frac{1}{\sqrt{2}}$$

Now we compute v_1 as

$$\begin{aligned} v_1 &= x_1 - \langle x_1, e_0 \rangle e_0 \\ &= t - \frac{1}{\sqrt{2}} \cdot \int_{-1}^1 t \cdot \frac{1}{\sqrt{2}} dt \\ &= t - \frac{1}{2} \int_{-1}^1 t dt \\ &= t - \frac{1}{2} \left[\frac{t^2}{2} \right]_{-1}^1 \\ &= t \end{aligned}$$

So the norm of v_1 is

$$\|v_1\| = \sqrt{\int_{-1}^1 t^2 dt} = \sqrt{\frac{2}{3}}$$

Then, the second element of the sequence is given by

$$e_1 = \frac{1}{\|v_1\|} v_1 = \sqrt{\frac{3}{2}} t$$

Finally, we compute v_2 as

$$\begin{aligned} v_2 &= x_2 - \langle x_2, e_1 \rangle e_1 - \langle x_2, e_0 \rangle e_0 \\ &= t^2 - \sqrt{\frac{3}{2}} t \cdot \int_{-1}^1 t^2 \cdot \sqrt{\frac{3}{2}} t dt - \frac{1}{\sqrt{2}} \cdot \int_{-1}^1 t^2 \cdot \frac{1}{\sqrt{2}} dt \\ &= t^2 - \frac{3}{2} t \cdot \int_{-1}^1 t^3 dt - \frac{1}{2} \int_{-1}^1 t^2 dt \\ &= t^2 - \frac{1}{3} \end{aligned}$$

Hence

$$\begin{aligned}\|v_2\| &= \sqrt{\int_{-1}^1 \left(t^2 - \frac{1}{3}\right)^2 dt} \\ &= \sqrt{\int_{-1}^1 \left(t^4 - \frac{2t^2}{3} + \frac{1}{9}\right) dt} \\ &= \sqrt{\frac{2}{5} - \frac{4}{9} + \frac{2}{9}} \\ &= \sqrt{\frac{8}{45}}\end{aligned}$$

so the third element of the sequence is

$$e_2 = \frac{1}{\|v_2\|} v_2 = \sqrt{\frac{45}{8}} \left(t^2 - \frac{1}{3}\right)$$

Just as a small sanity check we compute $\|e_2\|$ as follows

$$\begin{aligned}\|e_2\| &= \sqrt{\int_{-1}^1 \frac{45}{8} \left(t^2 - \frac{1}{3}\right)^2 dt} \\ &= \sqrt{\frac{45}{8}} \sqrt{\int_{-1}^1 \left(t^2 - \frac{1}{3}\right)^2 dt} \\ &= \sqrt{\frac{45}{8}} \|v_2\| \\ &= \sqrt{\frac{45}{8}} \sqrt{\frac{8}{45}} \\ &= 1\end{aligned}$$

□

3.5 - Series Related to Orthonormal Sequences and Sets

Proof. **1.** Suppose (6) converges to x i.e.

$$\sum_{k=1}^{\infty} \alpha_k e_k = x$$

Then by Theorem 3.5-2 we know that (7)

$$\sum_{k=1}^{\infty} |\alpha_k|^2$$

converges.

Let us take the squared norm of x then we have that

$$\|x\|^2 = \left\| \sum_{k=1}^{\infty} \alpha_k e_k \right\|^2$$

But since every $\alpha_k e_k$ is orthonormal then by the Pythagorean relation we have that

$$\left\| \sum_{k=1}^{\infty} \alpha_k e_k \right\|^2 = \sum_{k=1}^{\infty} \|\alpha_k e_k\|^2$$

Therefore

$$\|x\|^2 = \left\| \sum_{k=1}^{\infty} \alpha_k e_k \right\|^2 = \sum_{k=1}^{\infty} \|\alpha_k e_k\|^2 = \sum_{k=1}^{\infty} |\alpha_k|^2 \|e_k\|^2 = \sum_{k=1}^{\infty} |\alpha_k|^2$$

□

Proof. **3.** Let us consider the orthonormal sequence

$$e_1 = (0, 1, 0, 0, \dots)$$

$$e_2 = (0, 0, 1, 0, \dots)$$

$$e_3 = (0, 0, 0, 1, \dots)$$

$$\vdots$$

And let us consider the sequence

$$x = (1, 0, 0, 0, \dots) \in l^2$$

Then we see that

$$\sum_{k=1}^{\infty} \langle x, e_k \rangle e_k = (0, 0, 0, 0, \dots) \neq (1, 0, 0, 0, \dots) = x$$

□

Proof. 4.

Let $(x_j) \subset X$ such that $\|x_1\| + \|x_2\| + \dots$ converges.

We want to prove that (s_n) is Cauchy where $s_n = \sum_{i=1}^n x_i$. Taking $n', m' \in \mathbb{N}$ such that $n' < m'$, we see that

$$\left\| \sum_{i=1}^{n'} x_i - \sum_{j=1}^{m'} x_j \right\| = \left\| \sum_{i=n'+1}^{m'} x_i \right\| \leq \sum_{i=n'+1}^{m'} \|x_i\|$$

On the other hand, since $\sum_i \|x_i\|$ converges then, it's Cauchy, so, given $\epsilon > 0$ there is $N \in \mathbb{N}$ such that when $n, m \geq N$ we have that

$$\left| \sum_{i=1}^n \|x_i\| - \sum_{j=1}^m \|x_j\| \right| < \epsilon$$

We may assume that $n < m$, then

$$\left| \sum_{i=n+1}^m \|x_i\| \right| = \sum_{i=n+1}^m \|x_i\| < \epsilon$$

Hence, taking $n' = n$ and $m' = m$ we see that

$$\left\| \sum_{i=1}^n x_i - \sum_{j=1}^m x_j \right\| = \left\| \sum_{i=n+1}^m x_i \right\| \leq \sum_{i=n+1}^m \|x_i\| < \epsilon$$

Therefore (s_n) is Cauchy. □

Proof. 5.

From problem 4. we know that if X is an inner product space and $\sum_i \|x_i\|$ converges then the sequence (s_n) with $s_n = \sum_{i=1}^n x_i$ is Cauchy.

If instead we take (s_n) in a Hilbert space H , the space is complete so all Cauchy sequences converge, hence the sequence of partial sums (s_n) converge to some limit s , so we have that

$$\sum_{i=1}^{\infty} x_i = \lim_{n \rightarrow \infty} \sum_{i=1}^n x_i = \lim_{n \rightarrow \infty} s_n = s$$

Which implies that $\sum_{i=1}^{\infty} x_i$ converges. □

Proof. **6.**

Let (e_j) be an orthonormal sequence in Hilbert space H and let

$$x = \sum_{j=1}^{\infty} \alpha_j e_j \quad y = \sum_{j=1}^{\infty} \beta_j e_j$$

Let us define the partial sums sequences

$$x_n = \sum_{j=1}^n \alpha_j e_j \quad y_n = \sum_{j=1}^n \beta_j e_j$$

we see that $x_n \rightarrow x$ and $y_n \rightarrow y$ as $n \rightarrow \infty$.

Then

$$\langle x_n, y_n \rangle = \left\langle \sum_{j=1}^n \alpha_j e_j, \sum_{i=1}^n \beta_i e_i \right\rangle = \sum_{j=1}^n \alpha_j \sum_{i=1}^n \overline{\beta_i} \langle e_j, e_i \rangle = \sum_{j=1}^n \alpha_j \overline{\beta_j}$$

Because $x_n \rightarrow x$, $y_n \rightarrow y$ and the continuity of the inner product we have that $\langle x_n, y_n \rangle \rightarrow \langle x, y \rangle$. Therefore

$$\langle x, y \rangle = \lim_{n \rightarrow \infty} \langle x_n, y_n \rangle = \lim_{n \rightarrow \infty} \sum_{j=1}^n \alpha_j \overline{\beta_j} = \sum_{j=1}^{\infty} \alpha_j \overline{\beta_j}$$

□

Proof. 7.

Let (e_k) be an orthonormal sequence in a Hilbert space H and let $x \in H$. We want to show that

$$y = \sum_{k=1}^{\infty} \langle x, e_k \rangle e_k$$

exists in H and that $x - y$ is orthogonal to every e_k .

Let us consider the partial sum sequence

$$y_n = \sum_{k=1}^n \langle x, e_k \rangle e_k$$

If we let $m < n$ then we see that

$$\begin{aligned} \|y_n - y_m\|^2 &= \left\| \sum_{k=1}^n \langle x, e_k \rangle e_k - \sum_{j=1}^m \langle x, e_j \rangle e_j \right\|^2 \\ &= \left\| \sum_{k=m+1}^n \langle x, e_k \rangle e_k \right\|^2 \\ &= \sum_{k=m+1}^n \|\langle x, e_k \rangle e_k\|^2 \\ &= \sum_{k=m+1}^n |\langle x, e_k \rangle|^2 \end{aligned}$$

On the other hand, by Bessel inequality we know that $\sum_{k=1}^{\infty} |\langle x, e_k \rangle|^2$ converges so the tail of the series $\sum_{k=m+1}^n |\langle x, e_k \rangle|^2$ must tend to 0 as $n, m \rightarrow \infty$.

Therefore $\|y_n - y_m\|^2 \rightarrow 0$ as $n, m \rightarrow \infty$ and hence (y_n) is Cauchy and since $(y_n) \in H$ then (y_n) converges. So

$$y = \lim_{n \rightarrow \infty} \sum_{k=1}^n \langle x, e_k \rangle e_k = \sum_{k=1}^{\infty} \langle x, e_k \rangle e_k$$

exists in H .

Let us consider now the following

$$\begin{aligned} \langle x - y_n, e_k \rangle &= \left\langle x - \sum_{j=1}^n \langle x, e_j \rangle e_j, e_k \right\rangle \\ &= \langle x, e_k \rangle - \left\langle \sum_{j=1}^n \langle x, e_j \rangle e_j, e_k \right\rangle \\ &= \langle x, e_k \rangle - \sum_{j=1}^n \langle x, e_j \rangle \langle e_j, e_k \rangle \end{aligned}$$

Now taking the limit as $n \rightarrow \infty$ we get that

$$\begin{aligned}\lim_{n \rightarrow \infty} \langle x - y_n, e_k \rangle &= \langle x, e_k \rangle - \lim_{n \rightarrow \infty} \sum_{j=1}^n \langle x, e_j \rangle \langle e_j, e_k \rangle \\ \langle x - y, e_k \rangle &= \langle x, e_k \rangle - \sum_{j=1}^{\infty} \langle x, e_j \rangle \langle e_j, e_k \rangle \\ \langle x - y, e_k \rangle &= \langle x, e_k \rangle - \langle x, e_k \rangle \\ \langle x - y, e_k \rangle &= 0\end{aligned}$$

Where we used the continuity of the inner product, to show that as $n \rightarrow \infty$ we have that $\langle x - y_n, e_k \rangle \rightarrow \langle x - y, e_k \rangle$.

Therefore $x - y$ is orthogonal to every e_k .

$$\left\langle \sum_{j=1}^{\infty} \langle x, e_j \rangle e_j, e_k \right\rangle = \sum_{j=1}^{\infty} \langle x, e_j \rangle \langle e_j, e_k \rangle$$

□

Proof. **8.**

Let (e_k) be an orthonormal sequence in a Hilbert space H and let M be $M = \text{span}(e_k)$.

($\Rightarrow$) Let $x \in \bar{M}$ then x can be written as

$$x = \sum_{k=1}^{\infty} \alpha_k e_k$$

Where α_k are scalars. Since the summation converges to x then by Theorem 3.5-2 (b) we know the coefficients α_k are the Fourier coefficients $\langle x, e_k \rangle$ so x can be written as

$$x = \sum_{k=1}^{\infty} \langle x, e_k \rangle e_k$$

($\Leftarrow$) Let $x \in H$ such that it can be written as

$$x = \sum_{k=1}^{\infty} \langle x, e_k \rangle e_k = \sum_{k=1}^{\infty} \alpha_k e_k$$

Then x is an infinite linear combination of the elements of the orthonormal sequence (e_k) so must be that $x \in \bar{M}$.

□

Proof. 9.

Let (e_n) and $(\tilde{e}_n)$ be orthonormal sequences in a Hilbert space H and let $M_1 = \text{span}(e_n)$ and $M_2 = \text{span}(\tilde{e}_n)$.

($\Rightarrow$) Let $\bar{M}_1 = \bar{M}_2$. By problem 8 we know that if $x \in \bar{M}_1$ and $x \in H$ then it can be written as

$$x = \sum_{m=1}^{\infty} \alpha_m e_m = \sum_{m=1}^{\infty} \langle x, e_m \rangle e_m$$

We note that $e_n \in \bar{M}_2$ then we can write it as

$$e_n = \sum_{m=1}^{\infty} \langle e_n, \tilde{e}_m \rangle \tilde{e}_m = \sum_{m=1}^{\infty} \alpha_{nm} \tilde{e}_m$$

Where we defined $\alpha_{nm} = \langle e_n, \tilde{e}_m \rangle$.

In the same way, since $\tilde{e}_n \in M_1$ then we can write it as

$$\tilde{e}_n = \sum_{m=1}^{\infty} \langle \tilde{e}_n, e_m \rangle e_m = \sum_{m=1}^{\infty} \overline{\langle e_m, \tilde{e}_n \rangle} e_m = \sum_{m=1}^{\infty} \bar{\alpha}_{mn} e_m$$

($\Leftarrow$) Let us suppose that

$$e_n = \sum_{m=1}^{\infty} \alpha_{nm} \tilde{e}_m \quad \text{and} \quad \tilde{e}_n = \sum_{m=1}^{\infty} \bar{\alpha}_{mn} e_m$$

Where $\alpha_{nm} = \langle e_n, \tilde{e}_m \rangle$.

We see that e_n and $\tilde{e}_n$ can be written in the form of equation (6) then the orthonormal sequence (e_n) is both in $\bar{M}_1$ and $\bar{M}_2$ and the orthonormal sequence $(\tilde{e}_n)$ is also in $\bar{M}_1$ and $\bar{M}_2$.

Let us take $x \in \bar{M}_1$ then by problem 8 we know that x can be written as

$$x = \sum_{m=1}^{\infty} \langle x, e_m \rangle e_m$$

But since the orthonormal sequence (e_n) is also in $\bar{M}_2$ then this implies that $x \in \bar{M}_2$ and since x is arbitrary then

$$\bar{M}_1 \subset \bar{M}_2$$

The opposite can be proven if $x \in \bar{M}_2$ so we get that

$$\bar{M}_1 = \bar{M}_2$$

□

3.6 - Total Orthonormal Sets and Sequences

Proof. 1.

Let F be an orthonormal basis in an inner product space X . We know that

$$\overline{\text{span}(F)} = X$$

so if $x \in X$ can be represented as a linear combination of elements of F then $x \in \text{span}(F)$ but it could happen that $x \in X$ but it cannot be represented as a finite linear combination of elements in F .

For example, let us consider $X = l^2$ then if we take F as

$$F = \{(1, 0, 0, \dots), (0, 1, 0, \dots), (0, 0, 1, \dots), \dots\}$$

We see that $\overline{\text{span}(F)} = l^2$, also, the sequence $(1, 1/2, 1/3, \dots) \in l^2$ but is not in $\text{span}(F)$ since it's infinite. $\square$

Proof. 2.

Suppose the orthogonal dimension of a hilbert space H is finite, then there is a total orthonormal set T with finite number of elements in it, let us suppose this number is n , but by Lemma 3.4-2 we know that orthonormal sets are linealy independent so inside H we have a set of n linearly independent elements.

Let us suppose that there is, for example, one more element that is linearly independent with respect to the n previous elements, we want to arrive at a contradiction. So, in total we have $n + 1$ linearly independent elements, then, we can build an orthonormal set F of $n + 1$ elements using the Gram-Schmidt process, since this new orthonormal set was built from the original total orthonormal set T then also, must be that $\text{span}(F)$ is dense in H , but for this to happen by Lemma 3.3-7 must be that $\text{span}(F)^\perp = \{0\}$ which implies that $\text{span}(T)^\perp \neq \{0\}$ in the first place.

This is a contradiction and must be that there cannot be more than n linearly independent sets if the orthogonal dimension of H is n .

Therefore, the vector space H is of dimension n .

Now, suppose H is considered as a vector space is of dimension n (finite), then there is a linearly independent set with n elements in it. We can obtain an orthonormal set of n elements by applying the Gram-Schmidt process and we observe that all elements of H are in the span of the orthonormal set, so the orthonormal set is total, hence, the orthogonal dimension of H is n . $\square$

Proof. 4.

Let us consider first the case of a complex inner product space. From Parseval relation we note that

$$\|x - y\|^2 = \sum_k |\langle x - y, e_k \rangle|^2$$

Hence, from the LHS we see that

$$\|x - y\|^2 = \langle x - y, x - y \rangle = \langle x, x \rangle - \langle x, y \rangle - \langle y, x \rangle + \langle y, y \rangle$$

And, from the RHS we have that

$$\begin{aligned} \sum_k |\langle x - y, e_k \rangle|^2 &= \sum_k \langle x - y, e_k \rangle \overline{\langle x - y, e_k \rangle} \\ &= \sum_k (\langle x, e_k \rangle - \langle y, e_k \rangle) (\overline{\langle x, e_k \rangle} - \overline{\langle y, e_k \rangle}) \\ &= \sum_k \langle x, e_k \rangle \overline{\langle x, e_k \rangle} - \langle y, e_k \rangle \overline{\langle x, e_k \rangle} - \langle x, e_k \rangle \overline{\langle y, e_k \rangle} + \langle y, e_k \rangle \overline{\langle y, e_k \rangle} \\ &= \sum_k |\langle x, e_k \rangle|^2 - \langle y, e_k \rangle \overline{\langle x, e_k \rangle} - \langle x, e_k \rangle \overline{\langle y, e_k \rangle} + |\langle y, e_k \rangle|^2 \\ &= \langle x, x \rangle + \langle y, y \rangle - \sum_k \langle y, e_k \rangle \overline{\langle x, e_k \rangle} + \langle x, e_k \rangle \overline{\langle y, e_k \rangle} \end{aligned}$$

Joining both results we see that

$$\langle x, y \rangle + \langle y, x \rangle = \sum_k \langle y, e_k \rangle \overline{\langle x, e_k \rangle} + \langle x, e_k \rangle \overline{\langle y, e_k \rangle}$$

Now, let us consider

$$\|x - iy\|^2 = \sum_k |\langle x - iy, e_k \rangle|^2$$

From the LHS we see that

$$\begin{aligned} \|x - iy\|^2 &= \langle x - iy, x - iy \rangle \\ &= \langle x, x \rangle - \langle x, iy \rangle - \langle iy, x \rangle + \langle iy, iy \rangle \\ &= \langle x, x \rangle + i \langle x, y \rangle - i \langle y, x \rangle + \langle y, y \rangle \end{aligned}$$

And, from the RHS we have that

$$\begin{aligned} \sum_k |\langle x - iy, e_k \rangle|^2 &= \sum_k \langle x - iy, e_k \rangle \overline{\langle x - iy, e_k \rangle} \\ &= \sum_k (\langle x, e_k \rangle - i \langle y, e_k \rangle) (\overline{\langle x, e_k \rangle} + i \overline{\langle y, e_k \rangle}) \\ &= \sum_k \langle x, e_k \rangle \overline{\langle x, e_k \rangle} - i \langle y, e_k \rangle \overline{\langle x, e_k \rangle} + i \langle x, e_k \rangle \overline{\langle y, e_k \rangle} + \langle y, e_k \rangle \overline{\langle y, e_k \rangle} \\ &= \sum_k |\langle x, e_k \rangle|^2 - i \langle y, e_k \rangle \overline{\langle x, e_k \rangle} + i \langle x, e_k \rangle \overline{\langle y, e_k \rangle} + |\langle y, e_k \rangle|^2 \\ &= \langle x, x \rangle + \langle y, y \rangle + i \sum_k \langle x, e_k \rangle \overline{\langle y, e_k \rangle} - \langle y, e_k \rangle \overline{\langle x, e_k \rangle} \end{aligned}$$

Again, joining both results we see that

$$\langle x, y \rangle - \langle y, x \rangle = \sum_k \langle x, e_k \rangle \overline{\langle y, e_k \rangle} - \langle y, e_k \rangle \overline{\langle x, e_k \rangle}$$

Adding both result equations we get that

$$\begin{aligned} & \langle x, y \rangle + \langle y, x \rangle + \langle x, y \rangle - \langle y, x \rangle \\ &= \sum_k \langle y, e_k \rangle \overline{\langle x, e_k \rangle} + \langle x, e_k \rangle \overline{\langle y, e_k \rangle} + \sum_k \langle x, e_k \rangle \overline{\langle y, e_k \rangle} - \langle y, e_k \rangle \overline{\langle x, e_k \rangle} \end{aligned}$$

Hence

$$\langle x, y \rangle = \sum_k \langle x, e_k \rangle \overline{\langle y, e_k \rangle}$$

Let us consider now the case of a real inner product space. From the polarization identity we know that

$$\langle x, y \rangle = \frac{1}{2}(\|x\|^2 + \|y\|^2 - \|x - y\|^2)$$

Hence

$$\begin{aligned} \langle x, y \rangle &= \frac{1}{2}(\langle x, x \rangle + \langle y, y \rangle - \langle x, x \rangle - \langle y, y \rangle + \sum_k \langle y, e_k \rangle \langle x, e_k \rangle + \langle x, e_k \rangle \langle y, e_k \rangle) \\ &= \frac{1}{2}(2 \sum_k \langle x, e_k \rangle \langle y, e_k \rangle) \\ &= \sum_k \langle x, e_k \rangle \langle y, e_k \rangle \end{aligned}$$

Where we used the previous result we got for $\|x - y\|^2$ and that $\langle x, e_k \rangle = \langle e_k, x \rangle$ in a real inner product space. $\square$

Proof. 5.

Let (e_κ) , with $\kappa \in I$ be an orthonormal family in a Hilbert space H .

($\Rightarrow$) Let us suppose (e_κ) is total. Since Parseval relation

$$\sum_k |\langle x, e_k \rangle|^2 = \|x\|^2$$

Holds for any total orthonormal set M in a Hilbert space H and we derived the equation

$$\langle x, y \rangle = \sum_k \langle x, e_k \rangle \overline{\langle y, e_k \rangle}$$

Using the Parseval relation then it must also hold for the orthonormal family (e_κ) .

($\Leftarrow$) Let us suppose that for every $x, y \in H$ the following equation holds

$$\langle x, y \rangle = \sum_\kappa \langle x, e_\kappa \rangle \overline{\langle y, e_\kappa \rangle}$$

For the orthonormal family (e_κ) . Taking $y = x$ we get that

$$\|x\|^2 = \langle x, x \rangle = \sum_\kappa \langle x, e_\kappa \rangle \overline{\langle x, e_\kappa \rangle} = \sum_\kappa |\langle x, e_\kappa \rangle|^2$$

Then Parseval relation also holds and by Theorem 3.6-3 then must be that the orthonormal family (e_κ) is total.

□

Proof. 6.

Let H be a separable Hilbert space and $M = \{x_1, x_2, x_3, \dots\}$ a countable dense subset of H . Then, let us select

$$e_1 = \frac{x_1}{\|x_1\|}$$

To select e_2 we check if x_2 is in $\text{span}(e_1)$, if it is there then we drop it, if it's not then we define $v_2 = x_2 - \langle x_2, e_1 \rangle e_1$ and we select

$$e_2 = \frac{v_2}{\|v_2\|}$$

To select e_3 we check if x_3 is in $\text{span}(\{e_1, e_2\})$, if it is there then we drop it, if it's not then we define $v_3 = x_3 - \langle x_3, e_1 \rangle e_1 - \langle x_3, e_2 \rangle e_2$ and we select

$$e_3 = \frac{v_3}{\|v_3\|}$$

i.e. we follow the Gram-Schmidt process dropping some elements if necessary, to get an orthonormal sequence $(e_j) \subseteq M$.

Since the elements we dropped are in some span then they can be written as a linear combination of elements of (e_j) , so $M \subseteq \text{span}(e_j)$ but also we know that M is a dense subset of H so must be that $\text{span}(e_j)$ is also dense in H . Therefore (e_j) is a total orthonormal sequence obtained from M . $\square$

Proof. **7.**

Let H be a separable Hilbert space, then there is M a countable dense subset in H . As we showed before, using the Gram-Schmidt process we can find a total orthonormal sequence in H . $\square$

Proof. 8.

Let $F = (e_j)$ be an orthonormal sequence in a separable Hilbert space H . We know there exists $D = \{d_1, d_2, d_3, \dots\}$, a countable dense subset of H , because H is separable.

Let us take d_1 , if d_1 is linearly independent from the rest of the elements in D we keep it.

Next, we compute

$$v_1 = d_1 - \sum_{k=1}^{\infty} \langle d_1, e_k \rangle e_k$$

and

$$x_1 = \frac{v_1}{\|v_1\|}$$

i.e. we follow the Gram-Schmidt process to orthonormalize d_1 .

Finally, we add x_1 to a new orthogonal sequence $\tilde{F}$ defined as

$$\tilde{F} = \{x_1, x_2, \dots, e_1, e_2, \dots\}$$

To get each x_i we follow the same process for each d_i . If d_i is not linearly independent we drop it and continue.

Following this process we see that $\tilde{F}$ is an orthonormal sequence, and since we started from a countable dense subset D , then $\text{span}(\tilde{F})$ must also be dense in H , since we only dropped the elements that were linearly dependent to the other elements.

Therefore $\tilde{F}$ is a total orthonormal sequence that contains F . $\square$

Proof. **9.**

Let M be a total set in an inner product space X and let $\langle v, x \rangle = \langle w, x \rangle$ for all $x \in M$. We note that

$$\begin{aligned}\langle v, x \rangle &= \langle w, x \rangle \\ \langle v, x \rangle - \langle w, x \rangle &= 0 \\ \langle v - w, x \rangle &= 0\end{aligned}$$

This implies that $v - w \perp x$ but because Theorem 3.6-2, there is no element in X which is orthogonal to every element in M so must be that $v - w = 0$ and therefore $v = w$. $\square$

Proof. **10.**

Let M be a subset of a Hilbert space H and let $v, w \in H$. Suppose that $\langle v, x \rangle = \langle w, x \rangle$ for all $x \in M$ implies that $v = w$.

If M is not total then there is some $y \in H$, $y \neq 0$ such that $y \perp M$, we want to arrive at a contradiction.

Then $\langle y, x \rangle = 0$, but also we know that $\langle 0, x \rangle = 0$ so $\langle y, x \rangle = \langle 0, x \rangle$ hence $y = 0$ but y was different from 0, a contradiction.

Therefore M is total in H . □

3.7 - Legendre, Hermite and Laguerre Polynomials

Proof. 1.

The Legendre differential equation is

$$(1 - t^2)P_n'' - 2tP_n' + n(n + 1)P_n = 0$$

We note that applying the product rule to $[(1 - t^2)P_n']'$ gives us

$$[(1 - t^2)P_n']' = (1 - t^2)'P_n' + (1 - t^2)P_n'' = (1 - t^2)P_n'' - 2tP_n'$$

Then we can write the Legendre differential equation as

$$[(1 - t^2)P_n']' = -n(n + 1)P_n$$

Multiplying this equation by P_m , where $m \neq n$, gives us

$$P_m[(1 - t^2)P_n']' = -n(n + 1)P_nP_m$$

The corresponding equation for P_m multiplied by $-P_n$ is

$$-P_n[(1 - t^2)P_m']' = m(m + 1)P_mP_n$$

Then, adding the equation and integrating from -1 to 1 we see that

$$\begin{aligned} \int_{-1}^1 P_m[(1 - t^2)P_n']' - P_n[(1 - t^2)P_m']' dt &= \int_{-1}^1 m(m + 1)P_mP_n - n(n + 1)P_nP_m dt \\ &= [m(m + 1) - n(n + 1)] \int_{-1}^1 P_nP_m dt \end{aligned}$$

Let us integrate $\int_{-1}^1 P_m[(1 - t^2)P_n']' dt$ by parts

$$\begin{aligned} \int_{-1}^1 P_m[(1 - t^2)P_n']' dt &= P_m[(1 - t^2)P_n'] \Big|_{-1}^1 - \int_{-1}^1 P_m'[(1 - t^2)P_n'] dt \\ &= 0 - \int_{-1}^1 (1 - t^2)P_n'P_m' dt \end{aligned}$$

In the same way for $\int_{-1}^1 P_n[(1 - t^2)P_m']' dt$ we get that

$$\begin{aligned} \int_{-1}^1 P_n[(1 - t^2)P_m']' dt &= P_n[(1 - t^2)P_m'] \Big|_{-1}^1 - \int_{-1}^1 P_n'[(1 - t^2)P_m'] dt \\ &= 0 - \int_{-1}^1 (1 - t^2)P_n'P_m' dt \end{aligned}$$

Therefore

$$\begin{aligned} \int_{-1}^1 P_m[(1 - t^2)P_n']' - P_n[(1 - t^2)P_m']' dt &= - \int_{-1}^1 (1 - t^2)P_n'P_m' dt + \int_{-1}^1 (1 - t^2)P_n'P_m' dt \\ &= 0 \end{aligned}$$

So

$$\langle P_n, P_m \rangle = \int_{-1}^1 P_n P_m \, dt = 0$$

Hence (P_n) is an orthogonal sequence in the space $L^2[-1, 1]$.

□

Proof. 2.

The binomial theorem states that

$$(a + b)^n = \sum_{j=0}^n \frac{n!}{j!(n-j)!} a^{n-j} b^j$$

So in this case we can apply it to $(t^2 - 1)^n$ as follows

$$(t^2 - 1)^n = \sum_{j=0}^n \frac{n!}{j!(n-j)!} t^{2n-2j} (-1)^j$$

Let us take the n th derivative of t^{2n-2j} , we get that

$$\begin{aligned} \frac{d^n}{dt^n} t^{2n-2j} &= (2n-2j) \frac{d^{n-1}}{dt^{n-1}} t^{2n-2j-1} \\ &= (2n-2j) \cdot (2n-2j-1) \frac{d^{n-2}}{dt^{n-2}} t^{2n-2j-2} \\ &\vdots \\ &= [(2n-2j) \cdot (2n-2j-1) \cdot \dots \cdot (2n-2j-n)] t^{2n-2j-n} \\ &= \frac{(2n-2j)!}{(n-2j)!} t^{n-2j} \end{aligned}$$

Hence

$$\frac{d^n}{dt^n} (t^2 - 1)^n = \sum_{j=0}^n (-1)^j \frac{n!}{j!(n-j)!} \frac{(2n-2j)!}{(n-2j)!} t^{n-2j}$$

Therefore

$$\begin{aligned} P_n(t) &= \frac{1}{2^n n!} \frac{d^n}{dt^n} [(t^2 - 1)^n] \\ &= \frac{1}{2^n n!} \sum_{j=0}^n (-1)^j \frac{n!}{j!(n-j)!} \frac{(2n-2j)!}{(n-2j)!} t^{n-2j} \\ &= \sum_{j=0}^n (-1)^j \frac{(2n-2j)!}{2^n j!(n-j)!(n-2j)!} t^{n-2j} \end{aligned}$$

□

Proof. 3.

Let us consider the LHS as a binomial series

$$\frac{1}{\sqrt{1-2tw+w^2}} = (1-2tw+w^2)^{-1/2} = \sum_{k=0}^{\infty} \binom{-1/2}{k} (-2tw+w^2)^k$$

Now applying the binomial theorem we get that

$$\begin{aligned} \frac{1}{\sqrt{1-2tw+w^2}} &= \sum_{k=0}^{\infty} \binom{-1/2}{k} \sum_{j=0}^k \binom{k}{j} (-2tw)^{k-j} w^{2j} \\ &= \sum_{k=0}^{\infty} \binom{-1/2}{k} \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} 2^{k-j} t^{k-j} w^{2j+k-j} \\ &= \sum_{k=0}^{\infty} \sum_{j=0}^k (-1)^{k-j} \frac{(-1/2)(-1/2-1)\dots(-1/2-k+1)}{k!} \frac{k!}{j!(k-j)!} 2^{k-j} t^{k-j} w^{j+k} \\ &= \sum_{k=0}^{\infty} \sum_{j=0}^k (-1)^{k-j} \frac{2^k (-1/2)(-1/2-1)\dots(-1/2-k+1)}{2^k j!(k-j)!} 2^{k-j} t^{k-j} w^{j+k} \\ &= \sum_{k=0}^{\infty} \sum_{j=0}^k (-1)^{k-j} \frac{2^k \prod_{i=0}^{k-1} -(2i+1)/2}{2^j j!(k-j)!} t^{k-j} w^{j+k} \\ &= \sum_{k=0}^{\infty} \sum_{j=0}^k (-1)^{k-j} \frac{\prod_{i=0}^{k-1} -(2i+1)}{2^j j!(k-j)!} t^{k-j} w^{j+k} \\ &= \sum_{k=0}^{\infty} \sum_{j=0}^k (-1)^{k-j} \frac{(-1)^k \prod_{i=0}^{k-1} (2i+1)}{2^j j!(k-j)!} t^{k-j} w^{j+k} \\ &= \sum_{k=0}^{\infty} \sum_{j=0}^k (-1)^{2k-j} \frac{2k!}{2^{k+j} k! j!(k-j)!} t^{k-j} w^{j+k} \\ &= \sum_{k=0}^{\infty} \sum_{j=0}^k (-1)^j \frac{2k!}{2^{k+j} k! j!(k-j)!} t^{k-j} w^{j+k} \end{aligned}$$

Where we used in the last step that $(-1)^{2k-j} = (-1)^{2k}(-1)^{-j} = 1 \cdot (-1)^j$.

Applying the formula $\sum_{k=0}^{\infty} \sum_{j=0}^k C_{j,k} t^{k+j} = \sum_{k=0}^{\infty} \sum_{j=0}^{k/2} C_{j,k-j} t^k$ we get that

$$\begin{aligned} \frac{1}{\sqrt{1-2tw+w^2}} &= \sum_{k=0}^{\infty} \sum_{j=0}^{k/2} (-1)^j \frac{(2(k-j))!}{2^{k-j+j} (k-j)! j!(k-j-j)!} t^{(k-j)-j} w^{j+k-j} \\ &= \sum_{k=0}^{\infty} \sum_{j=0}^{k/2} (-1)^j \frac{(2k-2j)!}{2^k (k-j)! j!(k-2j)!} t^{k-2j} w^k \\ &= \sum_{k=0}^{\infty} P_k(t) w^k \end{aligned}$$



Proof. 4.

Let us consider

$$\frac{1}{r} = \frac{1}{\sqrt{r_1^2 + r_2^2 - 2r_1r_2 \cos \theta}}$$

but let us write it as

$$\frac{1}{\sqrt{r_1^2 + r_2^2 - 2r_1r_2 \cos \theta}} = \frac{1}{r_2} \frac{1}{\sqrt{1 + (r_1/r_2)^2 - 2(r_1/r_2) \cos \theta}}$$

Then taking $w = r_1/r_2$ and $t = \cos \theta$ the RHS has the form $1/r_2 \sqrt{1 - 2tw + w^2}$ and hence using the result of the previous problem we get that

$$\begin{aligned} & \frac{1}{r_2} \frac{1}{\sqrt{1 - 2(r_1/r_2) \cos \theta + (r_1/r_2)^2}} = \\ &= \frac{1}{r_2} \frac{1}{\sqrt{1 - 2wt + w^2}} \\ &= \frac{1}{r_2} \sum_{n=0}^{\infty} P_n(t) w^n \\ &= \frac{1}{r_2} \sum_{n=0}^{\infty} P_n(\cos \theta) \left(\frac{r_1}{r_2} \right)^n \end{aligned}$$

□

Proof. 5.

Let us consider $x(t) = \sum_{k=0}^{\infty} c_k t^k$, we note that

$$x(t)' = \sum_{k=1}^{\infty} c_k k t^{k-1}$$

$$x(t)'' = \sum_{k=2}^{\infty} c_k k(k-1) t^{k-2}$$

Replacing this into Legendre's differential equation gives us

$$(1-t^2)x(t)'' - 2tx(t)' + n(n+1)x(t) = 0$$

$$(1-t^2) \sum_{k=2}^{\infty} c_k k(k-1) t^{k-2} - 2t \sum_{k=1}^{\infty} c_k k t^{k-1} + n(n+1) \sum_{k=0}^{\infty} c_k t^k = 0$$

$$\sum_{k=2}^{\infty} c_k k(k-1) t^{k-2} - \sum_{k=2}^{\infty} c_k k(k-1) t^k - \sum_{k=1}^{\infty} 2c_k k t^k + \sum_{k=0}^{\infty} n(n+1) c_k t^k = 0$$

$$\sum_{k=0}^{\infty} c_{k+2}(k+1)(k+2) t^k - \sum_{k=2}^{\infty} c_k k(k-1) t^k - \sum_{k=1}^{\infty} 2c_k k t^k + \sum_{k=0}^{\infty} n(n+1) c_k t^k = 0$$

In the second term for $k = 0$ and 1 the terms are 0 , so we can set the sum to start at $k = 0$, the same applies for the third term, hence

$$\sum_{k=0}^{\infty} [c_{k+2}(k+1)(k+2) - c_k k(k-1) - 2c_k k + n(n+1)c_k] t^k = 0$$

Since the sum is equal to 0 then must be that each term is 0 , so we get that

$$c_{k+2}(k+1)(k+2) = [k(k-1) + 2k - n(n+1)]c_k$$

$$c_{k+2} = \frac{k(k-1) + 2k - n(n+1)}{(k+1)(k+2)} c_k$$

We see that

$$c_2 = -\frac{n(n+1)}{2} c_0 \quad k = 0$$

$$c_3 = \frac{2 - n(n+1)}{6} c_1 = -\frac{(n-1)(n+2)}{6} c_1 \quad k = 1$$

$$c_4 = \frac{6 - n(n+1)}{12} c_2 = -\frac{(n-2)(n+3)}{12} c_2 = \quad k = 2$$

$$= \frac{(n-2)(n+3)}{12} \frac{n(n+1)}{2} c_0 = \frac{(n-2)(n+3)n(n+1)}{24} c_0$$

...

Then we can write $x(t)$ as

$$x(t) = c_0 \left[1 - \frac{n(n+1)}{2!} t^2 + \frac{(n-2)(n+3)n(n+1)}{4!} t^4 + \dots \right] +$$

$$+ c_1 \left[t - \frac{(n-1)(n+2)}{3!} t^3 + \frac{(n-3)(n-1)(n+2)(n+4)}{5!} t^5 + \dots \right]$$

or defining $x_1(t)$ and $x_2(t)$ as

$$\begin{aligned} x_1(t) &= 1 - \frac{n(n+1)}{2!}t^2 + \frac{(n-2)(n+3)n(n+1)}{4!}t^4 + \dots \\ x_2(t) &= t - \frac{(n-1)(n+2)}{3!}t^3 + \frac{(n-3)(n-1)(n+2)(n+4)}{5!}t^5 + \dots \end{aligned}$$

We can write it as $x(t) = c_0x_1(t) + c_1x_2(t)$.

Finally, we note that when n is even, then the term containing t^{n+2} in $x_1(t)$ becomes 0, and since the coefficients are recursive, all subsequent coefficients become 0. The same thing happens when n is odd, so $x_2(t)$ becomes also a polynomial.

On the other hand, if we assume that

$$c_{n-2j} = (-1)^j \frac{(2n-2j)!}{2^n j! (n-j)! (n-2j)!}$$

Then $c_{n-2(j-1)} = c_{n-2j+2}$ must be

$$c_{n-2j+2} = (-1)^{j-1} \frac{(2n-2j+2)!}{2^n (j-1)! (n-j+1)! (n-2j+2)!}$$

But also by the recurrence relation, setting $k = n - 2j$ we have that

$$\begin{aligned} c_{n-2j+2} &= \frac{(n-2j)(n-2j-1) + 2(n-2j) - n(n+1)}{(n-2j+1)(n-2j+2)} c_{n-2j} \\ &= (-1)^j \frac{(n-2j)(n-2j-1) + 2(n-2j) - n(n+1)}{(n-2j+1)(n-2j+2)} \frac{(2n-2j)!}{2^n j! (n-j)! (n-2j)!} \\ &= (-1)^j [(n-2j)(n-2j+1) - n(n+1)] \frac{(2n-2j)!}{2^n j! (n-j)! (n-2j+2)!} \\ &= (-1)^j [n^2 - 2jn + n - 2jn + 4j^2 - 2j - n(n+1)] \frac{(2n-2j)!}{2^n j! (n-j)! (n-2j+2)!} \\ &= (-1)^j [-4jn + 4j^2 - 2j] \frac{(2n-2j)!}{2^n j! (n-j)! (n-2j+2)!} \\ &= (-1)^j [-2j(2n-2j+1)] \frac{(2n-2j)!}{2^n j! (n-j)! (n-2j+2)!} \\ &= (-1)^{j-1} \frac{2j!}{(j-1)!} \frac{(2n-2j+1)!}{2^n j! (n-j)! (n-2j+2)!} \\ &= (-1)^{j-1} \frac{2(n-j+1)}{(n-j+1)} \frac{(2n-2j+1)!}{2^n (j-1)! (n-j)! (n-2j+2)!} \\ &= (-1)^{j-1} \frac{(2n-2j+2)!}{2^n (j-1)! (n-j+1)! (n-2j+2)!} \end{aligned}$$

Therefore the coefficients of x_1 and x_2 match the coefficients of the Legendre polynomials, if we set correctly the c_0 and c_1 . $\square$

Proof. 6.

Let us consider the LHS as

$$e^{2wt-w^2} = \sum_{n=0}^{\infty} \frac{1}{n!} (2wt - w^2)^n$$

Applying the binomial theorem we get that

$$\begin{aligned} e^{2wt-w^2} &= \sum_{n=0}^{\infty} \frac{1}{n!} \sum_{k=0}^n \binom{n}{k} (2wt)^{n-k} (-w^2)^k \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} \sum_{j=0}^n \frac{n!}{j!(n-j)!} 2^{n-j} w^{n-j} t^{n-j} (-1)^j w^{2j} \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} \left[n! \sum_{j=0}^n (-1)^j \frac{2^{n-j}}{j!(n-j)!} t^{n-j} w^{n+j} \right] \end{aligned}$$

Also, we know that $\sum_{n=0}^{\infty} \sum_{j=0}^n C_{j,n} t^{n+j} = \sum_{n=0}^{\infty} \sum_{j=0}^{[n/2]} C_{j,n-j} t^n$.
Therefore

$$e^{2wt-w^2} = \sum_{n=0}^{\infty} \frac{1}{n!} \left[n! \sum_{j=0}^{[n/2]} (-1)^j \frac{2^{n-2j}}{j!(n-2j)!} t^{n-2j} w^n \right] = \sum_{n=0}^{\infty} \frac{1}{n!} H_n(t) w^n$$

□

Proof. 7.

We know that

$$H_n(t) = (-1)^n e^{t^2} \frac{d^n}{dt^n} (e^{-t^2})$$

So we see that

$$\begin{aligned} H'_n(t) &= (-1)^n \left[2te^{t^2} \frac{d^n}{dt^n} (e^{-t^2}) + e^{t^2} \frac{d^{n+1}}{dt^{n+1}} (e^{-t^2}) \right] \\ -H'_n(t) &= -2t(-1)^n e^{t^2} \frac{d^n}{dt^n} (e^{-t^2}) - (-1)^n e^{t^2} \frac{d^{n+1}}{dt^{n+1}} (e^{-t^2}) \\ -H'_n(t) &= -2t(-1)^n e^{t^2} \frac{d^n}{dt^n} (e^{-t^2}) + (-1)^{n+1} e^{t^2} \frac{d^{n+1}}{dt^{n+1}} (e^{-t^2}) \end{aligned}$$

Now, replacing the definition of $H_n(t)$ and that

$$H_{n+1}(t) = (-1)^{n+1} e^{t^2} \frac{d^{n+1}}{dt^{n+1}} (e^{-t^2})$$

We get that

$$-H'_n(t) = -2tH_n(t) + H_{n+1}(t)$$

Therefore

$$H_{n+1}(t) = 2tH_n(t) - H'_n(t)$$

□

Proof. **8.**

Let us differentiate the generating function as follows

$$\frac{d}{dt} \left(e^{2wt-w^2} \right) = 2we^{2wt-w^2} = \sum_{n=1}^{\infty} \frac{1}{n!} H'_n(t) w^n$$

The first term of the sum becomes 0 since $H_0(t) = 1$, so the sum starts at 1. Then we have that

$$\begin{aligned} 2w \sum_{n=0}^{\infty} \frac{1}{n!} H_n(t) w^n &= \sum_{n=1}^{\infty} \frac{1}{n!} H'_n(t) w^n \\ 2w \sum_{n=1}^{\infty} \frac{1}{(n-1)!} H_{n-1}(t) w^{n-1} &= \sum_{n=1}^{\infty} \frac{1}{n!} H'_n(t) w^n \\ 2 \sum_{n=1}^{\infty} \frac{1}{n!} n H_{n-1}(t) w^n &= \sum_{n=1}^{\infty} \frac{1}{n!} H'_n(t) w^n \\ \sum_{n=1}^{\infty} \frac{1}{n!} [2n H_{n-1}(t) - H'_n(t)] w^n &= 0 \end{aligned}$$

Where in the second step we changed the LHS sum to start from 1. Since the sum is equal to 0 then must be that each term is 0, hence

$$H'_n(t) = 2n H_{n-1}(t)$$

The Hermite differential equation states that

$$H''_n - 2t H'_n + 2n H_n = 0$$

From the equation we derived and the result of Problem 7 with $n-1$ we see that

$$H''_n = 2n H'_{n-1} = 2n(2t H_{n-1} - H_n)$$

Now, replacing in the Hermite differential equation we get that

$$\begin{aligned} H''_n - 2t H'_n + 2n H_n &= 2n(2t H_{n-1} - H_n) - 2t(2n H_{n-1}) + 2n H_n \\ &= 4nt H_{n-1} - 2n H_n - 4tn H_{n-1} + 2n H_n \\ &= 0 \end{aligned}$$

Therefore H_n satisfies the Hermite differential equation. □

Proof. **9.**

In the asymptotic case the differential equation $y'' + (2n + 1 - t^2)y = 0$ becomes $y'' = t^2 y$ for which we know that the solution is $y(t) = e^{-t^2/2}$ so we may assume that the solution to the original differential equation is of the form $y(t) = h(t)e^{-t^2/2}$, replacing this solution into the differential equation we have that

$$\begin{aligned} \frac{d^2 y}{dt^2} + (2n + 1 - t^2)y &= 0 \\ \frac{d^2}{dt^2} \left(h(t)e^{-t^2/2} \right) + (2n + 1 - t^2)h(t)e^{-t^2/2} &= 0 \\ e^{-t^2/2} (h(t)'' - 2th(t)' + (t^2 - 1)h(t)) + (2n + 1 - t^2)h(t)e^{-t^2/2} &= 0 \\ h(t)'' - 2th(t)' + (t^2 - 1)h(t) + 2nh(t) - (t^2 - 1)h(t) &= 0 \\ h(t)'' - 2th(t)' + 2nh(t) &= 0 \end{aligned}$$

We see that the differential equation is of the form of the Hermite differential equation, so H_n is a solution to the equation.

Therefore $H_n(t)e^{-t^2/2}$ is a solution to the original differential equation. $\square$

Proof. **10.**

Let us take the derivative of $e^{-t^2} H'_n$ as follows

$$(e^{-t^2} H'_n)' = -2te^{-t^2} H'_n + e^{-t^2} H''_n$$

Also, from Problem 8 we know that $H'_n = 2nH_{n-1}$ so

$$H''_n = 2nH'_{n-1} = 2n(2tH_{n-1} - H_n)$$

Where we used that $H'_{n-1} = 2tH_{n-1} - H_n$ from Problem 7.

Hence replacing

$$\begin{aligned} (e^{-t^2} H'_n)' &= -4nte^{-t^2} H_{n-1} + 4nte^{-t^2} H_{n-1} - 2ne^{-t^2} H_n \\ &= -2ne^{-t^2} H_n \end{aligned}$$

From the equation we just derived if we let $n \neq m$ and following the same method as in Problem 1, we can write that

$$\begin{aligned} H_m(e^{-t^2} H'_n)' - H_n(e^{-t^2} H'_m)' &= -2ne^{-t^2} H_n H_m + 2me^{-t^2} H_n H_m \\ &= 2e^{-t^2} (m - n) H_n H_m \end{aligned}$$

The (7a) functions state that

$$e_n = A_n e^{-t^2/2} H_n$$

where $A_n = 1/(2^n n! \sqrt{\pi})^{1/2}$.

We want to check that the functions are orthogonal on $\mathbb{R}$ so we need to integrate $e_n e_m$ from $-\infty$ to ∞

$$A_n A_m \int_{-\infty}^{\infty} e^{-t^2/2} H_n e^{-t^2/2} H_m dt = A_n A_m \int_{-\infty}^{\infty} e^{-t^2} H_n H_m dt$$

We note that this is the same as integrating $H_m(e^{-t^2} H'_n)' - H_n(e^{-t^2} H'_m)'$ times a multiplicative constant, so integrating the first term by parts we see that

$$\begin{aligned} \int_{-\infty}^{\infty} (e^{-t^2} H'_n)' H_m dt &= e^{-t^2} H'_n H_m \Big|_{t=-\infty}^{t=\infty} - \int_{-\infty}^{\infty} e^{-t^2} H'_n H'_m dt \\ &= 0 - \int_{-\infty}^{\infty} e^{-t^2} H'_n H'_m dt \end{aligned}$$

In the same way, integrating by parts the second term we get that

$$\begin{aligned} - \int_{-\infty}^{\infty} (e^{-t^2} H'_m)' H_n dt &= -e^{-t^2} H'_m H_n \Big|_{t=-\infty}^{t=\infty} + \int_{-\infty}^{\infty} e^{-t^2} H'_m H'_n dt \\ &= 0 + \int_{-\infty}^{\infty} e^{-t^2} H'_m H'_n dt \end{aligned}$$

Therefore

$$\int_{-\infty}^{\infty} H_m(e^{-t^2} H'_n)' - H_n(e^{-t^2} H'_m)' dt = 2(m-n) \int_{-\infty}^{\infty} e^{-t^2} H_n H_m dt = 0$$

Hence

$$\langle e_n, e_m \rangle = A_n A_m \int_{-\infty}^{\infty} e^{-t^2} H_n H_m dt = 0$$

Which impliest that the functions $e_n(t)$ are orthogonal. □

Proof. 11.

Let us consider the LHS of the equation as follows

$$\begin{aligned}\frac{1}{1-w}e^{-tw/(1-w)} &= \frac{1}{1-w} \sum_{n=0}^{\infty} \frac{1}{n!} \left(-\frac{tw}{1-w} \right)^n \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} \frac{(-1)^n w^n}{(1-w)^{n+1}} t^n\end{aligned}$$

Using the series expansion for $1/(1-w)^{n+1}$ we get that

$$\begin{aligned}\frac{1}{1-w}e^{-tw/(1-w)} &= \sum_{n=0}^{\infty} \frac{1}{n!} (-1)^n w^n t^n \sum_{k=0}^{\infty} \binom{k+n}{k} w^k \\ &= \sum_{n=0}^{\infty} \sum_{k=0}^{\infty} \frac{(-1)^n}{n!} \binom{k+n}{k} w^{n+k} t^n \\ &= \sum_{n=0}^{\infty} \sum_{j=n}^{\infty} \frac{(-1)^n}{n!} \binom{j}{n} w^j t^n\end{aligned}$$

Where in the last step we changed variables so $j = k + n$ and hence the sum starts at $j = n$.

If $n = 0$ then j can go from 0 to ∞ , if $n = 1$ then j can go from 1 to ∞ and so on. If we swap the summation signs for $j = 0$ then n can only be 0, for $j = 1$ then n can only be 0 or 1 and so on, then must be that

$$\frac{1}{1-w}e^{-tw/(1-w)} = \sum_{j=0}^{\infty} \sum_{n=0}^j \frac{(-1)^n}{n!} \binom{j}{n} w^j t^n$$

Or renaming variables $n \rightarrow j$ and $j \rightarrow n$ we get that

$$\frac{1}{1-w}e^{-tw/(1-w)} = \sum_{n=0}^{\infty} \sum_{j=0}^n \frac{(-1)^j}{j!} \binom{n}{j} t^j w^n$$

But from equation (10c) we know that

$$L_n(t) = \sum_{j=0}^n \frac{(-1)^j}{j!} \binom{n}{j} t^j$$

Therefore

$$\frac{1}{1-w}e^{-tw/(1-w)} = \sum_{n=0}^{\infty} L_n(t) w^n$$

□

Proof. **12.**

(a) Let us differentiate ψ with respect to w

$$\begin{aligned}\frac{d\psi(t, w)}{dw} &= \frac{d}{dw} \left(\frac{1}{1-w} e^{-tw/(1-w)} \right) \\ &= \frac{1}{(1-w)^2} e^{-tw/(1-w)} - \frac{1}{(1-w)} \frac{te^{-tw/(1-w)}}{(1-w)^2} \\ &= \frac{1}{1-w} \sum_{n=0}^{\infty} L_n w^n - \frac{t}{(1-w)^2} \sum_{n=0}^{\infty} L_n w^n\end{aligned}$$

From the RHS we have that

$$\frac{d}{dw} \left(\sum_{n=0}^{\infty} L_n w^n \right) = \sum_{n=1}^{\infty} n L_n w^{n-1} = \sum_{n=0}^{\infty} (n+1) L_{n+1} w^n$$

So

$$(1-w) \sum_{n=0}^{\infty} L_n w^n - t \sum_{n=0}^{\infty} L_n w^n = (1-w)^2 \sum_{n=0}^{\infty} (n+1) L_{n+1} w^n$$

Hence

$$\begin{aligned}\sum_{n=0}^{\infty} L_n w^n - \sum_{n=0}^{\infty} L_n w^{n+1} - t \sum_{n=0}^{\infty} L_n w^n &= \\ &= \sum_{n=0}^{\infty} (n+1) L_{n+1} w^n - 2 \sum_{n=0}^{\infty} (n+1) L_{n+1} w^{n+1} + \sum_{n=0}^{\infty} (n+1) L_{n+1} w^{n+2} \\ (1-t) \sum_{n=0}^{\infty} L_n w^n - \sum_{n=1}^{\infty} L_{n-1} w^n &= \\ &= \sum_{n=0}^{\infty} (n+1) L_{n+1} w^n - 2 \sum_{n=1}^{\infty} n L_n w^n + \sum_{n=2}^{\infty} (n-1) L_{n-1} w^n\end{aligned}$$

If we make every summation to start from $n = 2$ then

$$\begin{aligned}A + (1-t) \sum_{n=2}^{\infty} L_n w^n - \sum_{n=2}^{\infty} L_{n-1} w^n &= \\ &= B + \sum_{n=2}^{\infty} (n+1) L_{n+1} w^n - 2 \sum_{n=2}^{\infty} n L_n w^n + \sum_{n=2}^{\infty} (n-1) L_{n-1} w^n\end{aligned}$$

Where

$$\begin{aligned}A &= (1-t)[L_0 + L_1 w] - L_0 w \\ B &= L_1 + 2L_2 w - 2L_1 w\end{aligned}$$

Considering only what happens for $n \geq 2$ we can collapse the summations into one summation which is equal to 0, then must be that each term is equal to 0, so

$$\begin{aligned}(n+1)L_{n+1} - 2nL_n + (n-1)L_{n-1} - (1-t)L_n + L_{n-1} &= 0 \\(n+1)L_{n+1} - 2nL_n + nL_{n-1} - L_{n-1} - L_n + tL_n + L_{n-1} &= 0 \\(n+1)L_{n+1} - (2n+1-t)L_n + nL_{n-1} &= 0\end{aligned}$$

The last equation is the relation we are looking for.

(b) Let us differentiate ψ with respect to t

$$\begin{aligned}\frac{d\psi(t, w)}{dt} &= \frac{d}{dt} \left(\frac{1}{1-w} e^{-tw/(1-w)} \right) \\&= -\frac{we^{-tw/(1-w)}}{(1-w)^2} \\&= -\frac{w}{1-w} \sum_{n=0}^{\infty} L_n w^n\end{aligned}$$

From the RHS we have that

$$\frac{d}{dt} \left(\sum_{n=0}^{\infty} L_n w^n \right) = \sum_{n=1}^{\infty} L'_n w^n$$

Where we start from 1 now since $L_0 = 1$ then

$$\begin{aligned}-\frac{w}{1-w} \sum_{n=0}^{\infty} L_n w^n &= \sum_{n=1}^{\infty} L'_n w^n \\-w \sum_{n=0}^{\infty} L_n w^n &= (1-w) \sum_{n=1}^{\infty} L'_n w^n \\-\sum_{n=0}^{\infty} L_n w^{n+1} &= \sum_{n=1}^{\infty} L'_n w^n - \sum_{n=1}^{\infty} L'_n w^{n+1} \\-\sum_{n=1}^{\infty} L_{n-1} w^n &= \sum_{n=1}^{\infty} L'_n w^n - \sum_{n=2}^{\infty} L'_{n-1} w^n \\L_0 w + \sum_{n=2}^{\infty} L_{n-1} w^n &= -L'_1 w + \sum_{n=2}^{\infty} L'_{n-1} w^n - \sum_{n=2}^{\infty} L'_n w^n\end{aligned}$$

If we only consider the case when $n \geq 2$ then we get the following relation

$$L_{n-1} = L'_{n-1} - L'_n$$

□

Proof. 13.

Let us write equation (b) for $n + 1$, we see that

$$\begin{aligned} L_n(t) &= L'_n(t) - L'_{n+1}(t) \\ L'_{n+1}(t) &= L'_n(t) - L_n(t) \end{aligned}$$

Also, taking the derivative of equation (a) with respect to t we get that

$$(n + 1)L'_{n+1}(t) - (2n + 1)L'_n(t) + tL'_n(t) + L_n(t) + nL'_{n-1}(t) = 0$$

Replacing $L'_{n+1}(t)$ we get that

$$\begin{aligned} (n + 1)(L'_n(t) - L_n(t)) - (2n + 1)L'_n(t) + tL'_n(t) + L_n(t) + nL'_{n-1}(t) &= 0 \\ nL'_n(t) + L'_n(t) - nL_n(t) - L_n(t) - 2nL'_n(t) - L'_n(t) + tL'_n(t) + L_n(t) + nL'_{n-1}(t) &= 0 \\ -nL_n(t) - nL'_n(t) + tL'_n(t) + nL'_{n-1}(t) &= 0 \end{aligned}$$

Now, applying again equation (b) to replace $L'_{n-1}(t)$ we have that

$$\begin{aligned} -nL_n(t) - nL'_n(t) + tL'_n(t) + n(L_{n-1}(t) + L'_n(t)) &= 0 \\ -nL_n(t) + tL'_n(t) + nL_{n-1}(t) &= 0 \end{aligned}$$

Therefore

$$tL'_n(t) = nL_n(t) - nL_{n-1}(t) \quad (c)$$

We know that Laguerre's differential equation is

$$tL''_n + (1 - t)L'_n + nL_n = 0$$

Also, differentiating equation (c) with respect to t we have that

$$tL''_n + L'_n = nL'_n - nL'_{n-1}$$

Replacing this and equation (c), we see that

$$\begin{aligned} tL''_n + (1 - t)L'_n + nL_n &= nL'_n - nL'_{n-1} - nL_n + nL_{n-1} + nL_n \\ &= nL'_n - nL'_{n-1} + nL_{n-1} \\ &= nL'_n - nL'_{n-1} + n(L'_{n-1} - L'_n) \\ &= 0 \end{aligned}$$

Where we also replaced L_{n-1} with equation (b).

We see then that L_n satisfies Laguerre's differential equation. $\square$

Proof. 14.

Let us consider the functions

$$e_n(t) = e^{-t/2} L_n(t)$$

We want to show they have norm 1 so we need to compute $\|e_n\| = \sqrt{\langle e_n, e_n \rangle}$ as follows

$$\|e_n\|^2 = \int_0^\infty e^{-t} L_n(t) L_n(t) dt$$

We know that $L_n(t)$ is a polynomial of degree n , also, we know from Problem 15 that if $m < n$ then $\int_0^\infty e^{-t} t^m L_n(t) dt = 0$ so the only term surviving from $L_n(t)$ in our integral is the term of degree n i.e.

$$\|e_n\|^2 = \int_0^\infty e^{-t} L_n(t) L_n(t) dt = \alpha \int_0^\infty e^{-t} t^n L_n(t) dt$$

Where α is the coefficient of t^n . We can determine it using equation (10c), where we see that $\alpha = (-1)^n/n!$. Then

$$\begin{aligned} \|e_n\|^2 &= \frac{(-1)^n}{n!} \int_0^\infty e^{-t} t^n L_n(t) dt \\ &= \frac{(-1)^n}{n!} \int_0^\infty e^{-t} t^n \frac{e^t}{n!} f^{(n)}(t) dt \\ &= \frac{(-1)^n}{(n!)^2} \int_0^\infty t^n f^{(n)}(t) dt \\ &= \frac{(-1)^n}{(n!)^2} \left[t^n f^{(n-1)}(t) \Big|_0^\infty - n \int_0^\infty t^{n-1} f^{(n-1)}(t) dt \right] \\ &= \frac{n(-1)^{n+1}}{(n!)^2} \int_0^\infty t^{n-1} f^{(n-1)}(t) dt \end{aligned}$$

Where we used that $f(t) = t^n e^{-t}$ and we integrated by parts. So, integrating by parts n times we get that

$$\|e_n\|^2 = \frac{n!(-1)^{2n}}{(n!)^2} \int_0^\infty f^{(0)}(t) dt = \frac{1}{n!} \int_0^\infty t^n e^{-t} dt$$

We apply integration by parts again setting $u = t^n$ and $dv = e^{-t} dt$, hence

$$\|e_n\|^2 = \frac{1}{n!} \left[-t^n e^{-t} \Big|_0^\infty + n \int_0^\infty t^{n-1} e^{-t} dt \right] = \frac{n}{n!} \int_0^\infty t^{n-1} e^{-t} dt$$

Therefore, again applying integration by parts n times we get that

$$\|e_n\|^2 = \frac{n!}{n!} \int_0^\infty e^{-t} dt = 1$$

□

Proof. 15.

We want to show that the functions

$$e_n(t) = e^{-t/2} L_n(t)$$

constitute an orthogonal sequence in the space $L^2[0, +\infty)$.

First, we want to compute the integral $\int_0^\infty e^{-t} t^m L_n(t) dt$ where $m < n$.

Let us define $f(t) = t^n e^{-t}$ so $L_n(t) = e^t f^{(n)}(t)/n!$, then

$$\begin{aligned} \int_0^\infty e^{-t} t^m L_n(t) dt &= \frac{1}{n!} \int_0^\infty t^m f^{(n)}(t) dt \\ &= \frac{1}{n!} \left[t^m f^{(n-1)}(t) \Big|_0^\infty - \int_0^\infty m t^{m-1} f^{(n-1)}(t) dt \right] \\ &= -\frac{m}{n!} \int_0^\infty t^{m-1} f^{(n-1)}(t) dt \end{aligned}$$

Where we applied integration by parts and we used that $f^{(n)}(t)$ is of the form $f^{(n)}(t) = e^{-t} p_n(t)$ where $p_n(t)$ is a polynomial of degree n and when $t \rightarrow \infty$ we see that $f^{(n)}(t) \rightarrow 0$.

Applying integration by parts m times we get that

$$\begin{aligned} \int_0^\infty e^{-t} t^m L_n(t) dt &= (-1)^m \frac{m!}{n!} \int_0^\infty f^{(n-m)}(t) dt \\ &= (-1)^m (n-m)! \left[f^{(n-m-1)}(t) \right]_0^\infty \end{aligned}$$

We note that since $n-m-1 < n$ then the $n-m-1$ derivative of $f(t) = t^n e^{-t}$ is not going to have yet a constant term so we have that $f^{(n-m-1)}(t)|_{t=0} = 0$ and we know that $f^{(n-m-1)}(t) \rightarrow 0$ when $t \rightarrow \infty$, therefore

$$\int_0^\infty e^{-t} t^m L_n(t) dt = 0$$

If we take $L_m(t)$ with $m < n$ then $L_m(t)$ is a polynomial of degree smaller than n and hence

$$\langle e_n(t), e_m(t) \rangle = \int_0^\infty e^{-t} L_m(t) L_n(t) dt = 0$$

Therefore the functions $e_n(t)$ constitute an orthogonal sequence in the space $L^2[0, +\infty)$.

□